



Integrated PBPK modelling for PFOA exposure and risk assessment

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ABSTRACT

Per- and polyfluoroalkyl substances (PFASs) pose significant public health concerns due to their environmental persistence, bioaccumulation, and ubiquitous presence in human biomonitoring (HBM) data, despite regulatory restrictions. This study establishes a deterministic pharmacokinetic model for perfluorooctanoic acid (PFOA), enabling the estimation of PFOA concentrations in major human organs, even at low doses. The model integrates accumulation and recirculation mechanisms of PFOA in hepatic and renal tissues, leveraging publicly available HBM datasets (e.g., HBM4EU, NHANES, literature) to reconstruct bodyweight-normalized intake levels. Importantly, due to the extremely low urinary excretion concentrations of PFOA, most datasets were derived from blood-based measurements, particularly serum while confirming urine as unreliable biomarker of exposure. The analysis underscores the effectiveness of regulatory efforts in reducing PFOA exposures, as evidenced by declining time-trends in estimated exposure levels in recent studies. Risk characterization ratios were calculated based on recommended limits set by the European Food Safety Authority (EFSA), the United States, and Australia. While EFSA's tolerable weekly intake (TWI) indicated a high risk, other regulatory limits suggested less concern about risk at these intake levels. These findings highlight the need for continuous re-evaluation of exposures and targeted studies to identify key determinants of PFOA exposure, informing future regulatory measures. The study emphasizes the critical role of physiologically based pharmacokinetic (PBPK) modeling, HBM data, and exposure reconstruction in advancing chemical risk assessment. These tools form a science-based framework integral to the Chemical Strategy for Sustainability (CSS), enabling accurate predictions of internal exposure levels, empirical validation of models, and robust assessments of real-world exposure scenarios. The integration of these approaches supports the CSS goals of minimizing chemical risks while promoting innovation, ultimately contributing to a sustainable and protective regulatory landscape for human health and the environment.

1. Introduction

Perfluorooctanoic acid (PFOA) is an artificial compound that consists of an eight-carbon chain that is fully fluorinated, along with a functional group of carboxylic acid (Shu et al., 2023). PFOA belongs to a broad category of chemicals known as Per- and Polyfluoroalkyl Substances (PFAS), which are widely used for their water-, stain-, and oil-resistant properties as reported by Husøy et al. (2023). PFAS exhibit high

persistence due to their robust covalent carbon-fluorine (C–F) bonds, allowing them to bioaccumulate in the food chain and have a prolonged half-life in humans (Conder et al., 2008) and have been widely characterized as 'forever chemicals' due to their long half-lives in humans (Wee and Aris, 2023b). It has been reported that prenatal exposure to PFOA may influence the development of the fetus and child, potentially resulting in alterations in growth, behavior and learning as reported in human studies (Darrow et al., 2013; Johnson et al., 2014; Stein and

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Savitz, 2011; Stein et al., 2014). Importantly, the neurodevelopmental effects of PFOA have been demonstrated in numerous studies, particularly in mice (Goulding et al., 2017; Johansson et al., 2008; Mshaty et al., 2020; Wang et al., 2022). Other health effects associated with PFOA exposure include reproductive toxicity, as reported in human studies (Charazac et al., 2022; Di Nisio et al., 2019; Tarapore and Ouyang, 2021; Tsai et al., 2015) and rodent studies (Wang et al., 2021). Additionally, impaired immune system functionality has been observed in human studies (DeWitt et al., 2019; Grandjean et al., 2012), as well as in mice (Hu et al., 2012; Liang et al., 2022) and zebrafish (Liang et al., 2022). Elevated cholesterol levels have also been reported in human studies (Briksen et al., 2013; Fletcher et al., 2013; Fragki et al., 2021). Moreover, PFOA exposure has been linked to asthma in adolescents (Averina et al., 2019), obesity and hypertension in humans (Averina et al., 2021; Liu et al., 2018; Wee and Aris, 2023b), and an increased risk of cancer in humans (Barry et al., 2013; Bartell and Vieira, 2021). Last but not least, PFOA has been characterized as an endocrine disruptor as reported in the literature for humans (Berg et al., 2017; Steenland et al., 2015; Tarapore and Ouyang, 2021; Wee and Aris, 2023a, 2023b) as well as from in vitro studies (Di Nisio et al., 2020b).

Owing to its detrimental health impacts, PFOA has been phased out by major manufacturers since 2006 (Dassuncao et al., 2018; Xu et al., 2021a) and was restricted under the EU's Persistent Organic Pollutants Regulation (European Commission, 2023). The production and use of PFOA commenced in the 1950s, peaked between 1970 and 2002, and have subsequently decreased (Land et al., 2018; Prevedouros et al., 2006). PFOA exhibits high stability, resists degradation by oxidizing agents or strong acids, and is not susceptible to photolysis as reported by Buck and DeWitt (2015). Due to the fact that PFOA does not biodegrade in the environment (Prevedouros et al., 2006) and continues to persist globally (Olsen et al., 2012; Schröter-Kermani et al., 2013; Wu et al., 2017), its enduring presence remains a concern even after being phased out by numerous manufacturers (Wang et al., 2015). Characterized as both lipophobic and hydrophobic (Buck and DeWitt, 2015), it has been employed in the production of numerous consumer products, including pizza boxes, fast food wrappers, industrial lubricants, firefighting foams, nonstick cookware, surfactants, leather items, textiles, and stain-resistant coatings on carpets and fabrics (ATSDR, 2021). As a result of that, individuals can be exposed to PFOA through the consumption of contaminated food and water, as well as through contact with these materials (Paustenbach et al., 2007; Trudel et al., 2008; Wee and Aris, 2023a) resulting in PFOA being identified in the bodies (Berg et al., 2014; Kato et al., 2014; Liu et al., 2011; Mørck et al., 2015; Richterová et al., 2023; Schoeters et al., 2017). It is also important to note that PFOA can be encountered in airborne forms, further contributing to its exposure profile as reported by Chaemfa et al. (2010), Trudel et al. (2008), and Owens Jr (2021).

To achieve a comprehensive understanding of exposure to PFOA and other substances of concern, numerous human biomonitoring studies have been conducted at both national and international levels such as within the HBM4EU framework. These initiatives have produced an extensive repository of human biomonitoring (HBM) data, with meta-data systematically curated and made accessible via platforms such as IPCHEM and European HBM Dashboard. In addition to the numerous publications resulting from these projects (Bil et al., 2023; Katsonouri et al., 2023; Schillemans et al., 2023; Uhl et al., 2023), it is obvious that biomonitoring is increasingly becoming a central component of contemporary risk assessment (Karakoltzidis et al., 2025c; Sarigiannis et al., 2019). In addition to evaluating exposure levels within specific populations, HBM data can also be leveraged to estimate daily intakes by applying exposure reconstruction methodologies. Exposure reconstruction is a technique in environmental health and risk assessment that estimates the amount, duration, and frequency of past exposures to hazardous substances that an individual or population may have encountered. This process involves piecing together historical data, biological markers, and environmental measurements. PBPK models can

be employed to reconstruct exposure levels to a particular contaminant while other methodologies do exist such as e.g., personalized measurements and stochastic modelling (Karakoltzidis et al., 2025a). These applications offer the potential for a more precise understanding of health risks associated with past exposures, helping to evaluate the likelihood of adverse health outcomes and to inform regulatory or remedial actions (Georgopoulos et al., 2009; Sarigiannis & Karakitsios, 2019, 2021). Given that PBPK applications are well-suited for assessing exposure and human biomonitoring measurements can be utilized to estimate past exposure levels. It is important to note that there are already published studies related to PBPK modeling of PFOA (Choi et al., 2024; Fàbrega et al., 2014; Husøy et al., 2023; Lin et al., 2023; Loccisano et al., 2011; Vaccari et al., 2024; Worley et al., 2017b).

In the context of the present study, we utilize a generalized pharmacokinetic model for evaluating internal and external exposure. The current model is built upon the INTEGRA PBPK framework (Sarigiannis et al., 2014, 2016, 2020) and incorporates additional organs, including the pancreas, thyroid, thymus, and spleen, among others to conduct risk characterization. PFOA is recognized as a carcinogen (NIH, 2025) with significant impacts on numerous organs within the human body including kidney, liver, prostate, and thyroid. While its current effects may not be fully understood, ongoing research is expected to reveal more information regarding its modes of action across different organs. The data utilized in this study sourced from the IPCHEM and European HBM Dashboard platforms, NHANES studies, and literature. In light of this, a methodology for reconstructing distributions of human biomonitoring data is proposed when only aggregate data is available, effectively navigating concerns under the General Data Protection Regulation (GDPR (2016); Regulation (EU) 2016/679). The GDPR is a European Union legal framework governing the protection and privacy of personal data, which restricts the use of identifiable individual-level information. This method is applicable whenever there is interest in the overall variation of the dataset, rather than the exact values for each participant, thereby ensuring compliance with data privacy regulations. It addresses many challenges associated with data sharing and emphasizes the regulatory implications of such initiatives, providing an extensive application for utilizing HBM data. Recognizing the pressing need for pharmacokinetic applications, particularly those that can be rapidly implemented, the unfamiliarity within the scientific community regarding PBPK applications in risk assessment is addressed. To this end, conversion factors for intake estimation from serum, whole blood, and urine HBM data are developed.

Risk assessment, exposure analytics along with in silico methodologies are central components of the European Union's green transition, serving as fundamental tools to inform decision-making within computational pipelines supporting the advancement of Safe and Sustainable by Design (SSbD) initiatives and the Green Deal (Caldeira et al., 2022; Karakoltzidis et al., 2024, 2025b).

2. Methods

2.1. Software

Model code was written in R 4.2.2 (R Development Core Team, 2009) in a Windows 11 desktop PC. The PBPK model was developed with the employment of deSolve library (Soetaert et al., 2010). The graphs have been introduced using ggplot2 (Wickham, 2011). The boxplots employed in this study summarize the distribution of the data, displaying the median (center line), interquartile range (IQR; box height), and whiskers (extending to $1.5 \times$ IQR). Outliers were omitted for clarity. Skewed distributions were constructed with the VGAM package (Yee, 2008). Equations have been integrated into the graphs using the ggmisc library (Aphalo, 2016). Global sensitivity analysis output was performed using functionalities from pksensi package (Hsieh et al., 2020). Computational efficiency was enhanced through parallel processing using the parallel and foreach libraries (Eugster et al., 2011).

2.2. PBPK model development and structure

The PBPK model used in this study serves as the foundation for a robust and reliable risk assessment context. Consequently, there is a strong need for a concrete framework to assess exposure from PFOA in humans. INTEGRA software provides this framework by bringing in a source-to-dose continuum, environmental fate, exposure, and PBPK modelling over time, utilizing a Markov Chain Monte Carlo (MCMC) probabilistic analysis framework (Sarigiannis et al., 2014). The INTEGRA PBPK model is designed to comprehensively describe the ADME (absorption, distribution, metabolism, and excretion) processes in the human body across all life stages, enabling broad applicability to a wide range of chemicals with appropriate parameter adjustments. The current model version allows for the simulation of a three-step metabolic pathway, in addition to tracking the parent compound. Since PFOA is not metabolized (Steenland et al., 2010) only the parent compound module was employed.

The basis of the INTEGRA PBPK model construction is described in detail in the work of Sarigiannis et al. (2020). To further enhance the model, as described in previous works (Sarigiannis and Karakitsios, 2021; Sarigiannis et al., 2016, 2019), we included additional compartments such as the pancreas, thymus, spleen, and thyroid. Incorporating as many organs as possible into a PBPK model enhances physiological accuracy by capturing differences in blood flow, volume, metabolism, and chemical storage across organs. This approach allows the model to account for organ-specific toxic effects, such as those affecting the liver or kidneys, and reflects dynamic interactions between organs. To this end, a comprehensive literature review was conducted and is presented in Table 1, summarizing the impact of PFOA exposure across multiple human organs.

The methodology to increase the complexity of the equations for the additional compartments is described in detail in the work of Sarigiannis et al. (2016). Feces were included as an excretion compartment, with bile subsequently integrated as a subcompartment within the liver. The liver model was further divided into (a) red blood cells, (b) plasma and interstitial tissue, and (c) cellular components of the tissue. PFOA in feces was also enriched with the amount that was not absorbed in gastrointestinal tract (GI tract, digestive tract, alimentary canal). To more accurately represent the absorption dynamics of PFOA within the GI tract, two additional subcompartments were introduced: the stomach and the intestine. Moreover, each compartment is divided into three subcompartments, as described for the liver. The mass balance equation for each compartment accounts for all relevant values for parameters with biological significance, including absorption, metabolism, elimination, and protein binding (Sarigiannis et al., 2014, 2016, 2020). To account for the numerous cellular responses in the kidneys, the refined model incorporates multiple cell types, which are reflected in the model equations. Moreover, a filtrate compartment was considered in the construction of the model. The toxicokinetics of these processes are presented in detail in Section 2.4. All primary human organs have been considered in the model including arterial, venous, and portal blood compartments. The structure of the PFOA PBPK model is graphically presented in Fig. 1 while the mathematical model is provided in the supplementary material. Finally, it is assumed that all cells in the human body share a uniform spherical shape.

PFOA has the potential to enter the human body through multiple exposure routes, including oral, inhalation, and dermal pathways, as accounted for in the original INTEGRA PBPK model. However, in the present study, only oral exposure is considered. Additionally, it is important to acknowledge evidence indicating the loss of PFOA in women through menstrual blood (Upson et al., 2022), which was not incorporated into the development of the present PBPK model.

2.3. Physiological parameterization

Parameters pertaining to organ volumes (V) and blood flows (Q)

Table 1

Evidence to health impacts of PFOA to the human organs has been considered in the developing PBPK model.

Organ	Health Impact
GI Tract	Exposure to PFOA has been shown to exert pressure on the developing gut microbiota in humans (Gardner et al., 2020; Iszatt et al., 2019). In the maternal body, PFOA has been found to disrupt the homeostasis of infant gut microflora (Naspolini et al., 2022) and impair the human intestinal immune barrier (Zhang et al., 2024). Evidence from mice studies suggest that gestational exposure to PFOA induce maternal hepatic alterations through the gut-liver axis (Xu et al., 2022).
Liver	PFOA is known to cause liver damage, manifesting in numerous ways: (a) hepatomegaly, with evidence from both human and animal studies (Li et al., 2017); (b) liver enzyme alterations, supported by in vitro studies (Franco et al., 2020) and human data (Gallo et al., 2012; Lin et al., 2010); and (c) fatty liver disease, observed in humans (David et al., 2023; Wahlang et al., 2019) and mice (Sen et al., 2022). PFOA has also been shown to trigger inflammation, potentially leading to fibrosis in humans (Bassler et al., 2019), with additional in vitro evidence from liver cell lines (Qi et al., 2023). Chronic exposure to PFOA can exacerbate inflammation, accelerating the progression of liver damage in humans, as reported by Choi et al. (2022). Furthermore, PFOA induces oxidative stress, leading to cellular damage and further promoting fibrosis and liver injury, as supported by in vitro studies (Panaretakis et al., 2001). There is increasing evidence suggesting that prolonged exposure may elevate the risk of liver cancer, particularly hepatocellular carcinoma and cirrhosis in humans (Cao et al., 2022; Chang et al., 2014; Liu et al., 2021b; Yeung et al., 2013a). Epidemiological studies and animal models have highlighted a potential link between long-term PFOA exposure and liver tumor formation. Additionally, PFOA interferes with normal lipid metabolism in both humans (David et al., 2023; Wahlang et al., 2019) and mice (Sen et al., 2022), contributing to liver dysfunction, including cholesterol imbalance in humans (Andersen et al., 2021; Fragki et al., 2021) and mice (Das et al., 2017). It has also been shown to alter bile acid production, as observed in vitro cell lines (Behr et al., 2020; Louise et al., 2020), animal models (Wang et al., 2024b), and human studies (Salihović et al., 2020). Finally, PFOA has been widely recognized as an endocrine disruptor, with supporting evidence from human studies (Berg et al., 2017; Knox et al., 2011; Steenland et al., 2015; Tarapore and Ouyang, 2021; Wee and Aris, 2023a, 2023b) as well as in vitro research (Di Nisio et al., 2020b).
Pancreas	Chronic exposure to PFOA has been linked to pancreatic inflammation, as demonstrated in mouse studies (Kamendulis et al., 2022) and in vitro research (Abudayyak et al., 2021). This inflammation may impair the pancreas's ability to produce digestive enzymes, contributing to gastrointestinal issues, as highlighted by Hoyeck et al. (2022) in a comprehensive beta-cell review. Furthermore, PFOA disrupts endocrine functions, particularly by affecting insulin secretion from pancreatic beta cells. This disruption may contribute to insulin resistance, increasing the risk of metabolic disorders such as type 2 diabetes (Margolis and Sant, 2021; Niu et al., 2024; Qin et al., 2022; Rotondo and Chiarelli, 2020). Additionally, PFOA exposure has been associated with an elevated risk of pancreatic cancer, likely due to its ability to induce oxidative stress and inflammation, which disrupt normal cellular processes and promote tumor development. Evidence supporting this association has been observed in mice (Kamendulis et al., 2022), rats (Klaunig et al., 2012) and humans (Linja et al., 2024).
Spleen	Exposure to PFOA leads to immune system dysregulation in mice (Liang et al., 2022), zebrafish (Liang et al., 2022), and humans (Rudzanova et al., 2024). Ongoing research suggests that PFOA alters immune cell populations in the spleen, reducing the human body's ability to mount an adequate immune response (Taylor et al., 2023). Additionally, PFOA induces inflammation and oxidative stress, which damage spleen tissue and disrupt its normal functioning, as evidenced by in vitro studies (Zhang et al., 2014). These effects compromise the ability of human body to fight infections, increase susceptibility to autoimmune diseases, and may contribute to the development of lymphomas or other hematological disorders in humans (Dietert and Dietert, 2010; Semwal et al., 2022).
Thyroid	PFOA exposure interferes with the production and regulation of thyroid hormones, which is of significant importance in metabolism, growth, and development. Evidence from in vitro data (Coperchini et al., 2017, 2021), animal studies (Coperchini et al., 2017, 2021), and human cohort studies (Coperchini et al., 2017, 2021; Lee and Choi, 2017; Rattanukul, 2023; Wang et al., 2024c) supports this disruption. Specifically, PFOA exposure has been shown to decrease thyroid hormone levels in humans, particularly T4 (thyroxine), by impairing the thyroid gland's ability to synthesize and release these hormones (Coperchini et al., 2021). This

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Table 1 (continued)

Organ	Health Impact
Kidney	disturbance can lead to hypothyroidism, a condition characterized by fatigue, weight gain, and impaired metabolism (Coperchini et al., 2017; Lee and Choi, 2017). Additionally, PFOA may influence thyroid hormone transport by altering the binding proteins responsible for hormone regulation in the blood, as observed in humans (Coperchini et al., 2017; Duntas, 2023) and rat cell lines (Selano, 2018). Long-term exposure has also been associated with an increased risk of thyroid diseases, including thyroid cancer, due to its potential to induce cellular disruptions and promote inflammatory processes within the thyroid gland (Alsen et al., 2021; Coperchini et al., 2017; Liu et al., 2021a; Olsen et al., 2004). Exposure to PFOA leads to kidney damage by inducing oxidative stress and inflammation, which can impair the kidneys' ability to filter waste and regulate fluid balance, as reported in human studies (Bartell and Vieira, 2021; Liu et al., 2023; Steenland et al., 2010, 2022; Stevenson et al., 2021) and mouse studies (Rashid et al., 2020). There is evidence suggesting that PFOA exposure can cause structural alterations in kidney tissue, including fibrosis and nephropathy (Ferrari et al., 2021). These changes may progressively impair kidney function over time, increasing the risk of chronic kidney disease and other renal complications. Additionally, PFOA has been associated with an elevated risk of kidney cancer in humans, likely due to its ability to promote tumor development through mechanisms such as DNA damage, oxidative stress, and inflammation (Liu et al., 2023; Stevenson et al., 2021). Furthermore, PFOA affects kidney function by interfering with lipid metabolism and the renal excretion of other substances, potentially exacerbating kidney-related conditions (Chen et al., 2024a).
Adipose	PFOA exposure has the potential to alter adipocyte differentiation, as demonstrated in vitro studies (Modaresi et al., 2022; Reckziegel et al., 2024), potentially leading to increased fat accumulation and contributing to obesity. Evidence from human studies (Di Gregorio et al., 2019; Holtcamp, 2012) and additional in vitro research (Qin et al., 2022) supports its role in adipogenesis. Human studies suggest that PFOA interferes with lipid metabolism by disrupting the balance between fat storage and breakdown in adipose tissue, potentially leading to altered body fat distribution (La Merrill et al., 2013). Additionally, PFOA exposure has been associated with adipose tissue inflammation, a key factor in the development of metabolic disorders such as insulin resistance and type 2 diabetes, with supporting evidence from both human (Purdell et al., 2014; Timmermann et al., 2014) and animal studies (Purdell et al., 2014). Chronic exposure to PFOA may also impair endocrine function of adipose tissue, altering the secretion of adipokines—hormones that regulate appetite, metabolism, and inflammation in humans (Ding et al., 2021; Shelly et al., 2019). These metabolic disruptions may further increase the risk of conditions such as cardiovascular disease (Taube et al., 2012).
Bones	Ongoing research suggests that PFOA exposure in humans is able to disrupt bone remodeling by affecting the activity of osteoblasts and osteoclasts (Grammatiki et al., 2023; Hu et al., 2019; Kirk et al., 2021; Turan, 2021). This disruption may result in decreased bone mineral density and an increased risk of fractures, as bones become weaker and less resilient over time. Additionally, PFOA has been shown to interfere with the endocrine system, which is significantly important in regulating bone metabolism, particularly through hormones such as estrogen and parathyroid hormone (Iwobi and Sparks, 2023; Kirk et al., 2021; Park et al., 2024). This endocrine disruption may exacerbate conditions like osteoporosis, especially in postmenopausal women who are already at higher risk for bone loss (Priya et al., 2023). PFOA's potential to alter calcium and phosphate metabolism in bones further contributes to skeletal health concerns. Evidence from human studies (Baradaran Mahdavi et al., 2024; Koskela et al., 2017), animal models (Koskela et al., 2016), and in vitro research (Koskela et al., 2016, 2017) underscores the importance of understanding its long-term effects on bone integrity.
Brain	PFOA has been linked to multiple adverse effects on brain health, potentially impairing neurological function and increasing the risk of neurodegenerative disorders. Di Nisio et al. (2022) found evidence of these effects by analyzing brain autopsy samples from Italian residents in a PFAS-polluted area, which included exposure to PFOA. Prenatal and early-life exposure to PFOA interferes with hormonal signaling critical for brain development and function, potentially resulting in developmental delays, cognitive impairments, and behavioral changes (Bharal et al., 2024; Starnes et al., 2022; Yao et al., 2022). Human studies suggest that PFOA exposure induces oxidative stress and inflammation in human brain tissue, leading to neuronal damage and impairments in synaptic plasticity, which are essential for learning and memory (Bharal et al., 2024; Iqbal et al., 2020). PFOA has also been shown to cross the blood-brain barrier, allowing it to accumulate in the brain and disrupt

Table 1 (continued)

Organ	Health Impact
Gonads	cellular processes, including mitochondrial function and neurotransmitter balance (Guillotin and Delcourt, 2022). Chronic exposure to PFOA may increase the risk of neurodegenerative diseases, such as Alzheimer's and Parkinson's, by exacerbating mechanisms such as protein misfolding and neuroinflammation (Ashif et al., 2020; Pini et al., 2023). In males, PFOA exposure has been shown to disrupt testosterone production and sperm quality, leading to reduced fertility and altered reproductive function (Calvert et al., 2022; Shi et al., 2024; Tarapore and Ouyang, 2021). Ongoing research suggests that PFOA impairs testicular function, causing changes in hormone levels and sperm count, which may contribute to infertility or subfertility (Bach et al., 2016; Selvaraju et al., 2021). In females, PFOA exposure has been associated with disruptions in ovarian function, including altered levels of estrogen and progesterone—hormones essential for ovulation and menstrual regulation (Ding et al., 2020; Land et al., 2022). In vitro studies have also confirmed these effects (Ding et al., 2020). PFOA interferes with follicular development in the ovaries, potentially leading to ovulatory dysfunction and reduced fertility (Di Nisio et al., 2020a; Rickard et al., 2022). Furthermore, exposure to PFOA has been shown to affect the development of fetal gonads during pregnancy, as indicated by in vitro data from Zhou et al. (2022). This may have long-term reproductive health consequences for offspring (Kristensen et al., 2013).
Heart	Chronic exposure to PFOA has been linked to hypertension (Winquist and Steenland, 2014), a significant risk factor for heart disease and strokes. PFOA can disrupt lipid metabolism, leading to elevated cholesterol and triglyceride levels, which contribute to the formation of atherosclerotic plaques and an increased cardiovascular risk. Additionally, PFOA exposure is associated with altered heart rate variability, vasculitis, and endothelial dysfunction, further exacerbating cardiovascular health issues. These effects are supported by evidence from human cohorts (Boafo et al., 2023; Fragki et al., 2021; Huang et al., 2018; Lind and Lind, 2020; Mao et al., 2024; Meneguzzi et al., 2021; Schleizinger and Gokce, 2024; Shankar et al., 2012; Winquist and Steenland, 2014; Zhang et al., 2022).
Muscles	PFOA also induces inflammation that exacerbates fatigue and impairs muscle function, as demonstrated in vitro in cell lines (Solán et al., 2023). Additionally, it alters electrolyte and hormone balances, as shown in human studies (Wang et al., 2024a), which in turn affects muscle metabolism and performance.
Skin	PFOA negatively impacts skin health, contributing to conditions such as irritation, inflammation, and increased susceptibility to pathologies such as eczema and dermatitis, as evidenced by human studies (Fisal and Hassan, 2024; Lin et al., 2024; Mousavi et al., 2021). The oxidative stress and inflammation induced by PFOA is able to accelerate skin aging by breaking down collagen and elastin, which are significantly important for maintaining elasticity and firmness (Bertino et al., 2020; Grafanaki et al., 2023; Mousavi et al., 2021). Additionally, PFOA's ability to interfere with cellular processes and damage DNA could heighten the risk of nonmelanoma skin cancers, as noted by Moon and Mun (2024) based on NHANES datasets. Its endocrine-disrupting effects may also alter hormone regulation of sebum production and skin cell turnover, as demonstrated in in-vitro studies (Zhao et al., 2024).
Thymus	PFOA exposure weakens the immune system by impairing thymus function. Chronic exposure leads to thymic atrophy, reducing immune cell production and causing immunosuppression, which heightens susceptibility to infections and autoimmune diseases, as shown in animal studies (Liang et al., 2022). Additionally, in vitro studies suggest that PFOA induces inflammatory responses and oxidative stress, further damaging thymic tissue (Qazi et al., 2009).
Breast	As an endocrine disruptor, PFOA can interfere with hormonal balance, particularly by affecting estrogen regulation, which may disrupt breast cell growth and differentiation in females, as evidenced by rat studies and human cohorts (Kay et al., 2022). Chronic exposure may promote hormone-related breast cancer by altering estrogen receptor signaling and inducing oxidative stress and inflammation, both critical factors in cancer progression. This is strongly supported by in vitro studies (Pierozan et al., 2018, 2020) and human research (Lucaccioni et al., 2020). Additionally, PFOA exposure may increase the risk of benign conditions such as fibrocystic changes, complicating tumor detection and causing breast tenderness in humans (Siddique et al., 2016; Wielsøe et al., 2017).
Lung	PFOA exposure is associated with lung inflammation, as demonstrated in vitro and animal studies (Dragon et al., 2023), and oxidative stress, as shown in vitro research (Xin et al., 2018). It has been linked to an increased risk of respiratory conditions such as asthma and chronic

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Table 1 (continued)

Organ	Health Impact
	obstructive pulmonary disease (COPD) in humans (Mattila et al., 2021; Qin et al., 2017). PFOA may also disrupt immune responses in the lungs, increasing susceptibility to infections and diseases (Beans, 2021; Solan and Park, 2024). Additionally, PFOA exposure has the potential to impair pulmonary function by affecting surfactant production, which is important for lung inflation and preventing collapse, as evidenced in rat studies and in vitro data (Grasty, 2005; Sorli et al., 2020). Long-term exposure may raise the risk of lung cancer due to potential DNA damage and cellular mutations, with evidence drawn from both human and in vitro studies (Chen et al., 2024b; Jabeen et al., 2020).

were retrieved from the report of ICRP (2002). These parameters were then adjusted over time (t, in hours) to develop continuous, time-dependent nonlinear polynomial formulas. More information about that approach can be found in Sarigiannis et al. (2016, 2020). The equations for the organs that are not included in the original model were also retrieved from the report of ICRP (2002) and adjusted accordingly.

Table S1 provides the physiological parameters of the model that was not sourced from the ICRP document. These parameters were employed to optimize the functionalities of previously published INTEGRA PBPK models, facilitating the development of the PFOA model. The estimation of capillary surface areas relied on the study of Niederalt et al. (2018), using equation (9; herein will be referenced as 1), as well as the approach by Schmidt and Thews (2013). Additionally, the lifetime parameterization of the original INTEGRA PBPK model was considered in the present study. However, due to the lack of available information for non-adults, the model has not been validated for this population groups.

$$S_{organ} = k * f_{vas,organ} * V_{organ}$$
 (1)

Where S_{organ} is the capillary surface area, k is the proportionality constant, $f_{vas,organ}$ is the fraction of vascular space of an organ and V_{organ} is the organ volume. More information can be found in Niederalt et al. (2018).

2.4. PFOA toxicokinetics and ADME processes

This study builds on the INTEGRA PBPK model (Sarigiannis et al., 2016, 2020), extending it with additional compartments (e.g., thymus, pancreas, glomerular filtration, PFOA reabsorption) to improve internal exposure estimates. PFOA toxicokinetics involve complex absorption, distribution, and excretion processes. While exposure can occur via oral, dermal, and inhalation routes (Trudel et al., 2008), oral ingestion (e.g., contaminated food/water) is the primary pathway due to high bioavailability (Cui et al., 2022; EFSA et al., 2018). Dermal and inhalation exposures are minor, particularly in non-occupational settings. For this study, only oral exposure is considered.

After oral absorption, PFOA binds extensively to serum proteins (e.g., albumin), limiting its distribution and prolonging its half-life (Forsthuber et al., 2020; Salvalaglio et al., 2010; Wu et al., 2009). It distributes uniformly across tissues, with higher liver accumulation due to detoxification. Protein binding confines much of the compound to the bloodstream, resulting in a low volume of distribution (Worley et al., 2017b). PFOA undergoes no metabolic breakdown in humans due to its stable carbon-fluorine bonds (Conder et al., 2008), contributing to its persistence. Elimination occurs primarily via renal excretion (Han et al., 2012) and, to a lesser extent, fecal excretion (Worley et al., 2017b).

PFOA is primarily excreted renally, but its elimination is slow, resulting in a prolonged human half-life ranging from 0.5 to 15 years, depending on age and kidney function, whereas in rats, it varies between 2 h and 6 days (Andersen et al., 2006; Bartell et al., 2010; Brede et al., 2010; Butenhoff et al., 2004; Fasano et al., 2006; Li et al., 2022; Liu et al., 2023; Lou et al., 2009; Olsen et al., 2007; Tatum-Gibbs et al., 2011; Xu et al., 2020, 2021b; Yeung et al., 2013b; Zhang et al., 2013). Enterohepatic recirculation further delays excretion, as PFOA excreted in bile can be reabsorbed into the bloodstream. Additionally, maternal-fetal transfer (gestation/lactation) contributes to PFOA persistence (Blomberg et al., 2023; Liu et al., 2022), though these pathways are not considered in this study.

The model incorporates enterohepatic circulation, where PFOA cycles between the liver, bile, intestines, and bloodstream. After PFOA is

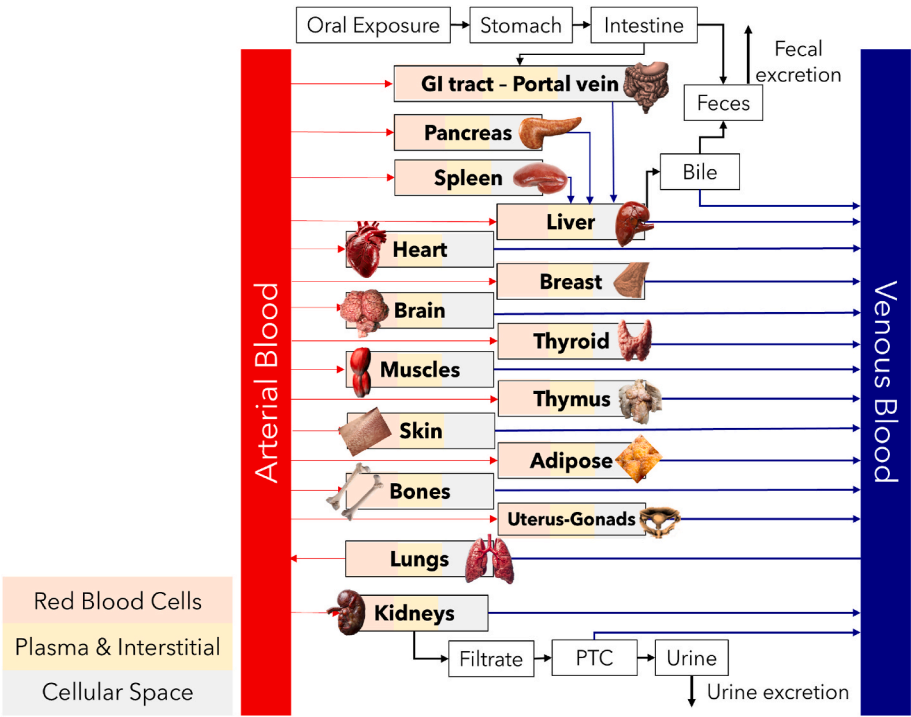


Fig. 1. Structure of the PBPK model employed. Refined from Sarigiannis et al. (2020). PTC term refers to proximal tubule cells. Not-colored compartments at the top of the figure, such as stomach and intestine only consisted of one equation.

excreted into the bile by the liver, it enters the intestinal tract where it can be reabsorbed back into the bloodstream, thus returning to systemic circulation. This process is facilitated by several transporters, including organic anion transporting polypeptides (OATPs: OATP1B1, OATP1B3, and OATP2B1) in the liver and sodium-dependent bile acid transporters (NTCP) in the intestine, which mediate the uptake and efflux of PFOA at multiple stages. In the intestinal lumen, PFOA can either be reabsorbed through enterocytes into the plasma or be excreted via feces. The dynamic equilibrium of PFOA among liver, bile, fecal, and plasma compartments is very important in determining its systemic bioavailability and prolonged biological persistence. This enterohepatic circulation pattern is particularly significant for hepatic accumulation, which directly contributes to the compound's potential to induce liver-specific adverse effects. The Michaelis-Menten kinetics for these transporters were parameterized using data from Lin et al. (2023). Unfortunately, no transporters beyond those listed above could be employed in the model development due to a lack of data (Vujic et al., 2024).

PFOA undergoes both glomerular filtration and tubular reabsorption in the kidneys which is governed by basolateral transporters (Liu et al., 2023). In the filtrate, it is either excreted in the urine or reabsorbed into the proximal tubule cells through active transport across apical membrane transporters (Niu et al., 2023; Ryu et al., 2024). Resorption can follow a Michaelis-Menten kinetic template as described by Worley et al. (2017b). Mass transport phenomena, including diffusion into and out of the proximal tubule cells and the efflux from these cells, are characterized by using first-order rate constants. Nakagawa et al. (2008) report that OAT1 (Slc22a6) and OAT3 (Slc22a8) facilitate the transport of PFOA across the basolateral membrane of proximal tubule cells (PTCs), promoting its excretion through the kidneys. On the other hand, URAT1 (urate transporter 1, Slc22a12) and OAT4 (Slc22a11) regulate the transport of PFOA across the apical membrane of proximal tubule cells, facilitating the reabsorption of PFOA into the bloodstream (Nakagawa et al., 2009; Yang et al., 2010).

A simplified depiction of the computational implementation of transporters in both the kidneys and liver within the model is illustrated in Fig. 2. Transporter kinetics (Km, Vmax, and in-vitro/in-vivo adjustment factors for each tissue) are presented in Table S2. In-vitro/in-vivo upscaling (or adjustment) factors for each transporter are treated as fitting parameters, used to translate in vitro data to human physiology. To that end, an adjustment factor was applied to the liver and kidneys to account for the upscaling from in vitro studies to humans. This approach

aligns with 3R principles and New Approach Methodologies (NAMs), supporting ethical science by reducing animal testing through in vitro-to-in vivo translation.

The kinetics of transporters were determined as outlined in Equation (2):

$$V = \frac{SF_{inVivo \rightarrow human\ liver/kidney} * V_{max,Transporter} * [C]}{[C] + Km_{Transporter}} \quad (2)$$

To improve model reliability, we implemented Fischer et al. (2024) equation (3) to estimate serum albumin-unbound PFOA fraction (fu), enhancing accuracy in bioavailability and toxicity predictions.

$$f_{unbound} = \frac{1}{1 + D_{HSA/W} * \frac{VF_{HSA}}{VF_W} + D_{glob/W} * \frac{VF_{glob}}{VF_W}} \quad (3)$$

Where VF_{HSA} (4.1 %), VF_{glob} (3.1 %), and VF_W (92.8 %) are the volume fractions of HSA, total globulins, and water in human serum (L L⁻¹) accordingly. D_{HSA/W} (10**4.48) and D_{glob/W} (10**2.16) are the partition coefficients of albumin and globulin in human serum respectively. The chemical specific parameters apart from organ: blood partition coefficients are provided in Table S3.

Partition coefficients were determined through comprehensive literature review. Where unavailable, a default value of 0.1 (representing rest-of-body, per Loccisano et al. (2011)) was applied and will be evaluated in sensitivity analysis. For compartments with multiple values, normal distributions were constructed for uncertainty analysis. The detailed workflow to estimate partition coefficients is presented in supplementary material and Tables S4 and S5. The aggregated values are presented in Table S6.

The estimation of permeability surface areas (PSAs) for each organ relied on the work of Niederalt et al. (2018). If hydraulic conductivity was not provided for the present model the value of muscle was employed. As carried out by the Niederalt et al. (2018), the estimation of chemical specific permeabilities for each organ is carried out separately. For this purpose, the following equation (4) was employed:

$$P_{Organ} = \frac{8 * \xi_{L,Organ} * D * a_{L,Organ} * \eta * Lp_{Organ}}{r_{L,Organ}^2} + \frac{8 * \xi_{S,Organ} * D * (1 - a_{L,Organ}) * \eta * Lp_{Organ}}{r_{S,Organ}^2} \quad (4)$$

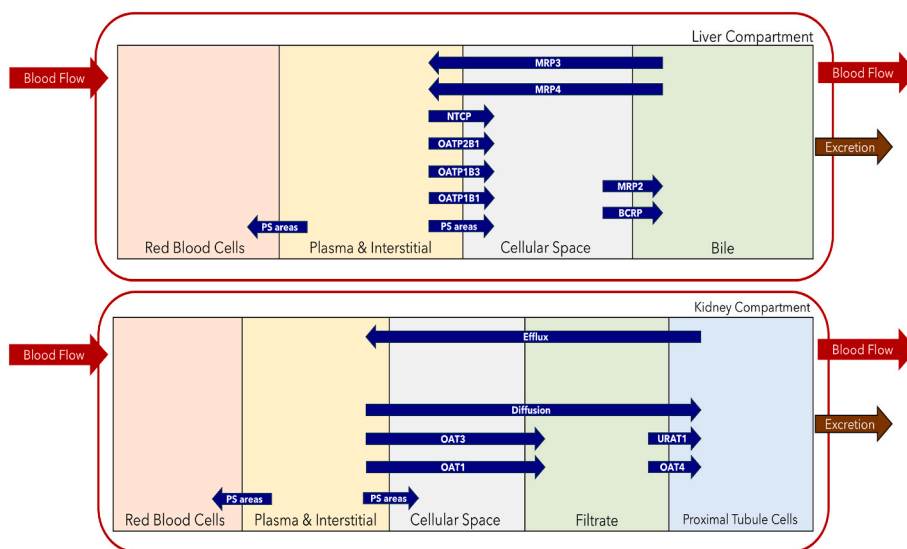


Fig. 2. Transport phenomena in the subcompartments of liver and kidneys as incorporated into the model. PS areas correspond to permeability-surface areas products. In the liver compartment, in-vitro kinetics were characterized for specific transporters: NTCP, OATP1B1, OATP1B3, and OATP2B1. For the remaining transporters, first-order kinetics were assumed.

Where s , L indicators are for small and large pores respectively, s_{organ} and L_{organ} are the small pores and large pores of each organ, ξ_{organ} is the dimensionless ratio of effective pore areas available for diffusion (through both small and large circular pores) to the total cross-sectional pore area, D is the diffusion coefficient of the solute, η is the viscosity of water in $1.17\text{E-}09\text{ N}\cdot\text{min}/\text{cm}^2$, r is the radius of small and large pores, and L_p is the hydraulic conductivity of the endothelium. It is important to highlight that the incorporation of PSAs allowed the consideration of passive diffusion across all tissues. The estimated permeabilities are provided in the supplementary material, Table S7.

2.5. Model calibration and evaluation

To enhance the validity of the PBPK model, an extensive literature review was conducted to identify studies measuring PFOA levels in the general population. A key inclusion criterion was the availability of repeated measurements in the same individuals over different time periods. The identified studies were categorized into two groups: the first group was used for model calibration, while the second group validated the model's generalizability across diverse studies. Calibration performed using data from Bartell et al. (2010) and ATSDR (2016) while it was evaluated with findings from Xu et al. (2020), Lahne et al. (2024), Perez et al. (2012) and Pérez et al. (2013). The daily water intake rate was set at 2.0 L/day, as recommended by EFSA for European populations (EFSA, 2010), and 1.36 L/day, based on EPA guidance for the US populations (USEPA, 2011).

The first study discussed is Bartell et al. (2010), which analysed serum samples collected at six time points from a cohort ($N = 40$) consuming water supplied by the Little Hocking Water Association or Lubeck Public Service District. Before the installation of the granular activated carbon filtration systems, four blood sampling campaigns were conducted; five, four, three, and two months prior to installation. The rest of the blood sampling campaigns were conducted for one and seven months after the installation. To fit the model to these serum PFOA levels, simulations were performed using both the maximum ($4.9\text{ }\mu\text{g/L}$) and the minimum ($1.9\text{ }\mu\text{g/L}$) drinking water concentrations. Following this, the drinking water concentration was set to zero to represent the installation of the granular activated carbon filtration systems. Similarly, regarding the individuals ($N = 132$) consumed water from the Lubeck Public Service District, seven blood sampling campaigns were taken place to measure serum PFOA levels. Almost one month before the installation of the granular activated carbon filtration systems, the first blood samples were collected. The remaining samples were collected after the installation at 1, 2, 3, 6, and 12 months. Again, two PFOA concentrations were used as inputs to the PBPK model: the lowest ($0.41\text{ }\mu\text{g/L}$) and the highest ($1.0\text{ }\mu\text{g/L}$). Afterward, the drinking water input in the model was set to zero. In both simulations, the assumption from Fromme et al. (2009) was applied, setting non-drinking water exposure levels for PFOA at $0.01\text{ }\mu\text{g/h}$ until the filters be installed. Subsequently, it was set to zero $\mu\text{g/h}$. Our model incorporated a 30-year pre-filter exposure period, based on the assumption that pseudo-steady state conditions would be achieved following this duration of continuous exposure. This study was initially employed to adjust predicted levels of PFOA in serum to the ones measured by Bartell et al. (2010).

The second study identified originates from the ATSDR (2016) and was utilized for model fitting as well. The model was calibrated using serum PFOA concentrations measured in individuals residing in North Alabama in 2010 and 2016, who reported consuming contaminated drinking water supplied by the West Morgan East Lawrence Municipal Water Authority. According to Fromme et al. (2009), non-drinking water exposure levels for PFOA were estimated at approximately $0.01\text{ }\mu\text{g/h}$ which after the installation of granular filtration were set to zero $\mu\text{g/h}$. The PFOA concentration in drinking water was assumed to be $0.04\text{ }\mu\text{g/L}$, reflecting the average level reported in treated water supplied by the municipal authority during that period. Active exposure ceased in

2016 after the installation of granular activated carbon filtration systems. Blood samples from this community were collected in April 2010 and between January and February 2016. Morning urine voids were also collected during the second sampling period; however, the ATSDR (2016) and Worley et al. (2017a) documents only reported median and mean values for the total number of participants. A 30-year exposure window was assumed before the installation of the filtration systems. Model predictions were simultaneously tested to the ATSDR data as well as with those from Bartell et al. (2010) to calibrate urine data parameters and validate the model's accuracy in maintaining a strong fit to serum concentrations. Exposure scenarios incorporated normal distributions with a mean daily urinary void frequency of $7 (\pm 2\text{ standard deviations})$.

In a separate study conducted by Xu et al. (2020), it was found that employees at a municipal airport in northern Sweden had been exposed to elevated levels of PFAS, including PFOA, through their drinking water. The groundwater in the area was contaminated by firefighting foam containing high concentrations of PFAS, a common issue near airports. The study aimed to characterize the PFAS profile in drinking water and biological samples (paired serum and first morning urine) while also estimating the serum half-lives of measured PFAS. Blood samples were obtained from all 26 airport employees within two weeks of their transition to clean water, with 17 individuals subsequently monitored on a monthly basis over a five-month period. Additionally, Xu et al. (2020) collected first morning voids samples to extend their analysis which is something that allow for the evaluation of the model's applicability to urinary PFOA levels. However, as the results were provided as specific gravity-adjusted values without the corresponding aggregated data, the methodology outlined by Graul and Stanley (1982) to normalize the urine data was utilized. This approach assumed a normal distribution of specific gravities with a minimum value of 1.01 g/cm^3 , a maximum value of 1.03 g/cm^3 , a mean value of 1.024 g/cm^3 , and a standard deviation of $\pm 0.005\text{ g/cm}^3$. Last but not least utilizing the aggregated data provided in the work of Xu et al. (2020), an optimized distribution was developed to be compared with the model outputs. As suggested by Xu et al. (2020) the exclusive exposure pathway of the airport employees is through the PFOA-contaminated water and consequently the daily intake rate is estimated $0.6\text{ }\mu\text{g/day}$. For model evaluation purposes, we assume daily urinary void frequency follows a normal distribution with a mean of seven voids/day and standard deviation of two voids/day.

Lahne et al. (2024) reported that PFOA was used as an emulsifier in fluoropolymer production at a plant located in the Altötting district of southern Bavaria, Germany. This industrial activity resulted in the direct release of PFOA into the environment, contaminating local drinking water supplies. During a human biomonitoring survey conducted in 2018, residents of the area exhibited elevated median PFOA serum levels compared to a control group from Munich, Germany, where no known PFOA exposure occurred. A follow-up study aimed to evaluate the effectiveness of drinking water purification, the primary intervention to reduce PFOA exposure, and to identify factors influencing PFOA levels. Only participants from the original 2018 HBM study were included. Blood samples were collected between June and August 2022, with 764 individuals participating in the follow-up study, including the general population, women of reproductive age, and children under 12 years old. The results showed a significant reduction in median PFOA serum levels across all groups within the general population. Additionally, age was identified as a key predictor of internal PFOA exposure, highlighting the importance of developing lifetime exposure models to better assess PFAS exposure. Lahne et al. (2024) reported PFOA concentrations ranging from 0.004 to $0.18\text{ }\mu\text{g/L}$. The model was evaluated using both concentrations.

Perez et al. (2012) developed a method to quantify PFAS, including PFOA, in human urine and hair, which they applied to participants in Barcelona, Spain. Urine measurements are regarded as a random urine spot for each participant. However, their study did not report

participants' consumption patterns as well as physiological characteristics. To address the exposure gap, we conducted a literature review to estimate PFOA exposure sources for the Spanish population. Dietary intake data from Pérez et al. (2014) indicated mean daily PFOA intakes of 60 ng/kg/day for males and 49 ng/kg/day for females. The value of males was tested in the evaluation process while the bodyweight was assumed to be 70 kg. Drinking water concentrations in Barcelona, sourced from Ericson et al. (2009), ranged from 0.85 ng/L (minimum) to 57.43 ng/L (maximum). We simulated exposure scenarios using these values, incorporating normal distribution with a mean daily urinary void frequency of 7 (± 2 standard deviations). While Perez et al. (2012) also measured PFOA concentrations in hair, these pieces of information were excluded from our analysis as they fall outside the scope of this study.

Pérez et al. (2013) analysed the concentrations of 21 PFASs in 99 autopsy tissue samples, including brain, liver, lung, bone, and kidney tissues, from individuals residing in Tarragona, Spain. PFASs were detected in all tissue types, with distinct accumulation patterns observed depending on the tissue. These findings are particularly significant for the validation of PBPK models currently being developed for humans. However, Pérez et al. (2013) declare that further research is needed to investigate the distribution of PFASs throughout the human body in greater detail. Fàbrega et al. (2014) estimated a daily PFOA intake of 1.55 ng/kg bodyweight/day, representing 98 % of total exposure, i.e., 0.11 $\mu\text{g/day}$.

2.6. Sensitivity analysis/uncertainty analysis

The global sensitivity analysis (GSA) in this study was rigorously designed to assess the influence of model parameters on output variability while explicitly accounting for their joint distributions and interactions. Unlike local sensitivity analyses—which assess parameters individually by perturbing one variable at a time around a fixed nominal value; the present GSA approach simultaneously varies all parameters across their physiologically plausible ranges. This underscores why an extensive literature review was conducted to identify as many chemical-specific parameters as possible. The methodology ensured that nonlinear relationships and interdependencies between parameters were captured, a critical requirement for robust uncertainty quantification in complex PBPK models. To implement the GSA, physiological and chemical-specific parameters (Table S8; supplementary material) were assigned in normal distributions reflecting their inherent uncertainty. While normal distributions were selected for computational tractability, the standard deviation of each chemical-specific parameter was carefully defined as 10 % of its mean value. If minimum and maximum values were available for a parameter (e.g., for most partition coefficients), these values were incorporated into the distribution construction setting up upper and lower limits. With regard to physiological parameters, a 30 % standard deviation was selected in order to capture as much as possible the dynamics of heterogeneous populations. Physiological parametrization relied on ICRP (2002) as documented previously.

The analysis utilized Sobol's variance-based sensitivity method; a gold-standard GSA technique implemented via the pksensi R package. Total order indices decompose the total variance in model outputs into contributions from individual parameters (first-order effects) and parameter interactions (higher-order effects). This approach explicitly quantifies how synergies or antagonisms between parameters—such as those governing PFOA absorption, distribution, and clearance—collectively influence predictions. Parallel computing was employed within pksensi to execute thousands of iterative simulations, enabling rapid aggregation of results for sensitivity index calculations and accommodating the long duration of each simulation (>30 years). Data schemas were structured to align with the pksensi framework, ensuring compatibility with its built-in functions for variance decomposition.

While GSA does not reduce uncertainty; which stems from limited empirical data (e.g., unknown population-average transporter expression levels or unmeasured inter-individual variability); it elucidates the relative contributions of uncertain parameters to model output variability. GSA provides a systematic way to prioritize which uncertainties most significantly impact predictions, guiding future data collection efforts to reduce uncertainty empirically.

2.7. Exposure reconstruction

Exposure reconstruction, or reverse dosimetry, represents another essential tool in the arena of risk assessment, enabling the estimation of external exposures by integrating probabilistic modelling with HBM data. This technique involves running PBPK models in reverse to approximate intakes and assessing the risks associated with exposures. In this study, a Bayesian Markov Chain Monte Carlo (MCMC) framework was incorporated into the PBPK model employing the reconstruction of external exposure scenarios. This methodology builds upon previous works by Georgopoulos et al. (2009) and Sarigiannis et al. (2019). The Bayesian MCMC approach enables probabilistic estimation of exposure parameters by fitting the PBPK model to observed biomarker concentrations. Here, biomarker concentrations in whole blood, serum, and urine were measured. Additionally, the intake was modelled as being received on a continuous daily basis throughout the simulations. While this approach does not capture day-to-day variation in real-world exposure patterns, it provides a reasonable approximation for PFOA due to its long biological half-life, which dampens the impact of short-term fluctuations on long-term body and tissue levels. The analysis incorporated the assumption that individual daily void frequencies are normally distributed (mean $\pm$ SD: 7 ± 2 voids/day) across the study population. Consequently, the model running backward fitted on the mean concentration of number of spots each time. This probabilistic framework also accounts for variability in physiological characteristics (e.g., body weight, metabolism rates) and parameter uncertainty, ensuring comprehensive and reliable assessments. Posterior distributions were sampled using MCMC methods, providing convergence diagnostics and credible interval estimations. The reconstructed exposure estimates serve as a fundamental foundation for the risk characterization ratios as they will be detailed later on. Such approaches have proved particularly valuable when direct measurements of external exposures are unavailable, simultaneously enhancing the transparency and reliability of exposure evaluations. Finally, the approximate intakes are employed to estimate conversion factors for all matrices.

2.8. Human biomonitoring (HBM) data

When developing the PBPK model for reverse dosimetry estimation, careful consideration was given to the selection of HBM data to ensure robustness and relevance. In addition to that, HBM is significantly important for the EU's Chemical Strategy for Sustainability and Green Deal, as it provides real-world insights into population exposure to hazardous chemicals, enabling evidence-based policies to prioritize harmful substances and safeguard human health (Karakoltzidis et al., 2025c). Initially, it is important to note that the selection of HBM data was exclusively based on American and European populations. This focus is largely due to the availability of extensive and detailed data on these populations, which share common physiological characteristics. Moreover, the physiological attributes of these groups are well-documented, with comprehensive stratification by factors such as age and sex (ICRP, 2002), allowing for the inclusion of interindividual variability in the PBPK model development.

To collect HBM data, we consulted three sources. First, we utilized the aggregated data provided by the European HBM Dashboard for PFOA across all human matrices. The European HBM Dashboard offers comprehensive aggregate data on human exposure to environmental chemicals across Europe. The data from nearly every study includes

selected percentiles (P05, P10, P25, P50, P75, P90, and P95), number of participants, bioequivalent (BE) values, HBM guidance values, and biomonitoring guidance values. The data sourced from the Dashboard is illustrated in Fig. 3. Participant ages were randomly sampled between study-specific minimum and maximum values when exact ages were unavailable.

Secondly, data from NHANES studies was utilized to estimate daily intake levels for U.S. citizens. The NHANES data, obtained from the CDC website (CDC, 2025), is presented in Fig. 4. The NHANES program provides comprehensive datasets for each cycle of data collection, with over 2,000 participants included in each cycle. These datasets offer detailed information on PFOA exposures, in contrast to the HBM4EU dashboard, which provides only aggregated data.

NHANES applies a unique sampling approach to ensure representation of populations frequently overlooked or inadequately included in epidemiological research. Its study designs deliberately overrepresents specific demographic groups such as African Americans, Hispanic communities, adolescents, and older adults in order to address gaps in common study populations. The strategy involves a stratified, multi-stage framework, including 15 US counties selected annually as primary sampling units. Due to this complex structure, researchers analyzing NHANES data usually apply specialized survey sampling techniques to account for weighting and clustering, ensuring accurate national estimates rather than relying on standard statistical methods (Ale et al., 2024). However, this falls outside the scope of the current study, which specifically reconstructed PFOA intakes based exclusively on serum concentrations and age. To that end, the participants' body-weight was also not considered and estimated intakes were normalized according to the mean value of ~80 kg. Participants without PFOA-serum concentration were excluded from the rest of the analysis.

Additional relevant studies were identified through comprehensive literature review. Seven HBM studies were identified in the literature, including those by Falandysz et al. (2006), Midasch et al. (2006), Rylander et al. (2009), Hölzer et al. (2008), De Felip et al. (2015),

Thépaut et al. (2021), and Hartmann et al. (2017). The results of this review are summarized and visualized in Fig. 5. Since studies provided only age ranges, we assigned participant ages by sampling between each study's minimum and maximum values.

Falandysz et al. (2006) measured whole blood concentrations of multiple PFAS, including PFOA, in 15 participants from the Gulf of Gdańsk region in Poland. Midasch et al. (2006) assessed plasma concentrations of PFOS and PFOA in 105 non-smoking individuals from the German general population, providing an initial estimate of the exposure levels in Germany. The median plasma concentrations across all participants were 6.8 mg/L for PFOA, with the 95th percentiles at 14.6 mg/L. Rylander et al. (2009) investigated dietary intake, age, gender, and body mass index as potential predictors of PFAS exposure in a study population from northern Norway, comprising 44 women and 16 men. In another study conducted by Hölzer et al. (2008), 40,000 residents in Arnsberg, Germany, were exposed to drinking water contaminated with PFAS. The cross-sectional study included 170 children (ages 5–6), 317 mothers (ages 23–49), and 204 men (ages 18–69). De Felip et al. (2015) measured PFOA concentrations in serum samples collected between 2011 and 2012 from 549 Italian women of reproductive age. For the conversions of the ng/g (to µg/L) values reported in this study, the blood density value provided in Table 2 was applied. Thépaut et al. (2021) analysed PFAS concentrations in 144 EuroMix participants (44 males, 100 females), examining associations with diet and personal care product use.

To reconstruct distributions of the data from each study retrieved (HBM4EU dashboard; Literature), it was assumed that the mean value for each study could be estimated using equation (5), and the standard deviation using relation (6) if these pieces of information were not available. This approach enabled the introduction of a randomly skewed distribution with several observations corresponding to the number of participants reported in each HBM study. Since variance must be positive, the shape of the distribution was determined using relation (7), provided that the P25, P50, and P75 percentiles were available. If any of

Human Biomonitoring of PFOA: HBM4EU dashboard

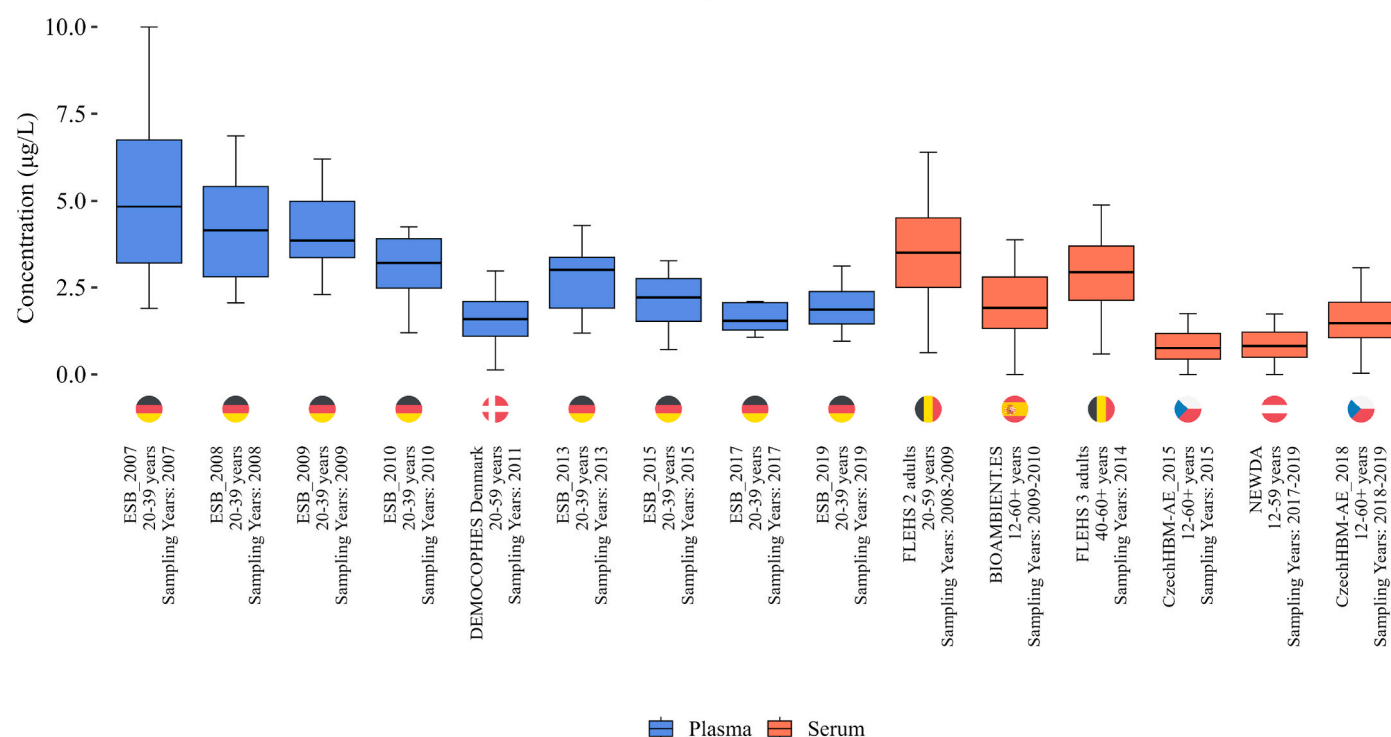


Fig. 3. Presentation of HBM data on PFOA concentrations derived from European HBM Dashboard.

Human Biomonitoring of PFOA: NHANES data

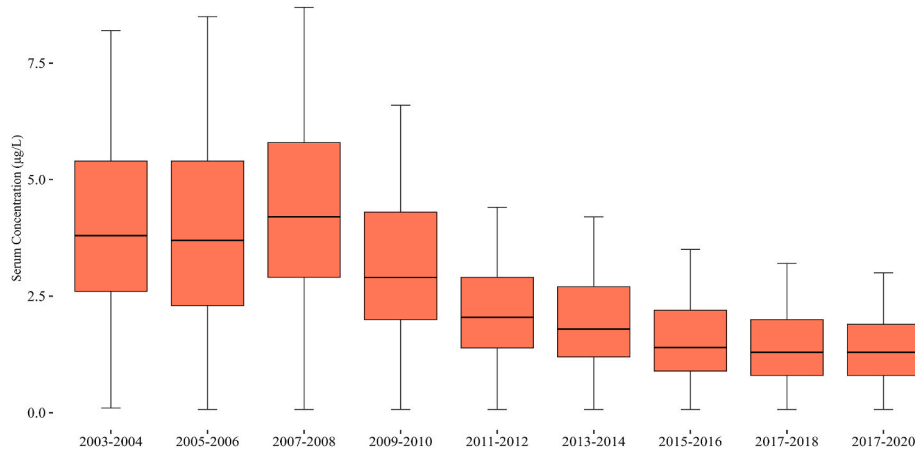


Fig. 4. Presentation of NHANES HBM data for PFOA concentrations.

Human Biomonitoring of PFOA: Literature data

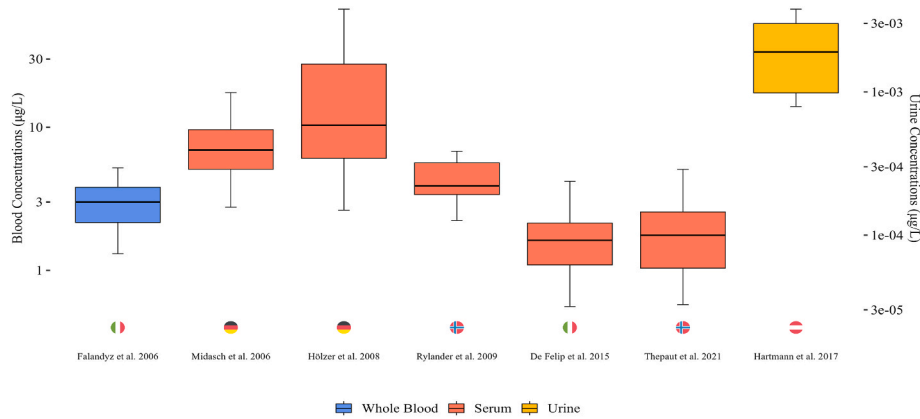


Fig. 5. The data derived from extensive literature research is presented here. For illustration purposes, the data is displayed on a log10 scale. The studies referenced originate from the works of Falandysz et al. (2006), Midasch et al. (2006), Rylander et al. (2009), Hölzer et al. (2008), De Felip et al. (2015), Thépaut et al. (2021), and Hartmann et al. (2017).

Table 2

Presentation of TDI/TWI of multiple safety authorities worldwide as well as the BEs based on these values.

Organization	TDI/TWI	Unit	$BE_{Serum} \left(\frac{\mu g}{L} \right)$	$BE_{Whole\ Blood} \left(\frac{\mu g}{L} \right)$	$BE_{Urine} \left(\frac{\mu g}{L} \right)$
EFSA (2008a)	1.5	$\frac{\mu g}{kg_{BW}day}$	1359	725	5.90 ± 1.25
EFSA (2020)	0.0044	$\frac{\mu g}{kg_{BW}week}$	0.56	0.29	0.001 ± 0.0004
US EPA (2017)	0.02	$\frac{\mu g}{kg_{BW}day}$	20.20	10.77	0.05 ± 0.004
FSANZ (2021)	0.16	$\frac{\mu g}{kg_{BW}day}$	141.70	73.60	0.50 ± 0.018

these percentiles were below the limit of quantification, the shape was set to one. The constructed distribution was subsequently optimized to align its percentiles with those of the reference distribution. The synthetic distribution ranged from zero to $1.5 \times P95$, if max value was not provided, closely matching the actual distribution's descriptive statistics.

$$\mu \approx \frac{\sum_{i=1}^{N_{percentiles}} P_i}{N_{percentiles}} \quad (5)$$

$$\sigma \approx \sqrt{\frac{\sum_{i=1}^{N_{percentiles}} (P_i - \mu)^2}{N_{percentiles}}} \quad (6)$$

$$shape \approx \frac{P75 + P25 - 2P50}{P75 - P25} \quad (7)$$

where $i = 1$ represents P05, $i = 2$ represents P10, $i = 3$ represents P25, $i = 4$ represents P50, and so on.

The NHANES data was provided in its entirety, so no imputation

algorithms were necessary. For the imputation of the literature data, a similar approach was applied. In contrast to the methodology used for the data from the HBM4EU dashboard, the studies identified in the literature also provided mean values and standard deviations, along with selected percentiles, which facilitated the imputation of artificial PFOA concentrations in serum.

2.9. Risk characterization

Risk characterization of PFOA was conducted using distinct methodologies that incorporated both external and internal exposure estimates. The methodology applied in the present work relies on the one suggested by Sarigiannis et al. (2019). Specifically, intake estimates were directly compared to tolerable daily intakes (TDIs) provided in Table 2, as determined by multiple safety authorities worldwide. Additionally, biomonitoring equivalents (BEs) for serum, whole blood and urinary data were approximated based on each TDI and are presented in Table 2. The derivation of these BEs assumed a continuous dose equivalent to the TDI for an individual weighing 70 kg, as recommended by Sarigiannis et al. (2019). For the US population, we adopted 80 kg as the representative average body weight, based on comprehensive data from NHANES studies. Accordingly, the corresponding dosimetry estimates (TDI/TWI) were applied to the fitted PBPK model in a forward manner, which was run for three years to estimate the bioequivalent concentrations of PFOA in serum and urine (morning void). Finally, it is important to highlight that the Tolerable Weekly Intake (TWI) established by EFSA has been considered as being equally distributed across continuous seven-day exposure events and exclusively originated from exposure to PFOA.

European Food Safety Authority (EFSA); United States Environmental Protection Agency Health Advisory Board (US EPA); Food Standards Australia-New Zealand (FSANZ)

The substantial reduction in EFSA's tolerable intake for PFOA, from 1.5 µg/kg body weight/day (TDI; 2008) to 0.0044 µg/kg body weight/week (TWI; 2020), reflects a paradigm shift in scientific understanding and regulatory risk assessment over this period. This 2,400-fold difference arises from advances in toxicological research, recognition of PFOA's unique bioaccumulative properties, and evolving methodologies to protect vulnerable populations (EFSA, 2020). The 2008 TDI was primarily based on animal studies identifying liver and kidney toxicity as critical effects, with safety margins derived from high-dose experiments in male rats (EFSA, 2008b). By contrast, the 2020 TWI incorporated human epidemiological evidence demonstrating harm at far lower exposures, particularly for sensitive endpoints such as developmental immunotoxicity (e.g., reduced vaccine efficacy in children) and hormonal disruption. These effects, observed in populations with background-level PFOA exposure, revealed risks previously underestimated. Furthermore, EFSA's 2020 assessment prioritized PFOA's extreme persistence in the human body, necessitating a focus on cumulative, lifelong exposure rather than short-term effects. However, efforts are ongoing in this direction through collaboration with European-level regulatory offices and EU member states via the 'PFAS Initiative Group' which aims to share information and develop collaborative approaches for PFAS risk assessment (EFSA, 2024).

Urine is not a suitable matrix for assessing human PFOA exposure, with serum being the preferred biomarker. Due to PFOA's minimal excretion rate and the high variability in daily urine voids, urinary concentrations are often too low for reliable detection, as evidenced by the scarcity of such data in literature. Additionally, daily void frequency significantly impacts morning concentration levels. For these reasons, Table 2 presents distributions of urine concentrations as BE values rather than exact measurements, and serum is strongly recommended for accurate exposure assessment. Nevertheless, this study's methodological framework enables estimation of intake levels and risk ratios from urine data. The derived mean and standard deviation for urinary BEs assume a population void frequency of 7 ± 2 voids per day. Given PFOA's long

half-life, this assumption introduces temporal variations, justifying the use of a concentration distribution rather than single values. However, risk ratios are calculated exclusively using mean concentrations.

3. Results and discussion

3.1. Model validation

3.1.1. Model calibration

The first dataset used to calibrate the model sourced from the work of Bartell et al. (2010) who studied serum PFOA levels in two cohorts ($N = 40$ and $N = 132$) consuming water from districts in Little Hocking and Lubeck before and after installing granular activated carbon (GAC) filtration. Blood samples were collected at intervals ranging from five months pre-installation to 12 months post-installation. PBPK model simulations used maximum/minimum drinking water PFOA concentrations (4.9/1.9 µg/L for Little Hocking; 1.0/0.41 µg/L for Lubeck) and later set concentrations to 0 µg/h to model post-filtration effects. The aim of this fitting process was first to adjust the predicted serum PFOA levels to the ones reported by Bartell et al. (2010). Predictions against serum PFOA measurements are presented in Fig. 6. Simulation time was 33 years.

The PBPK model was then calibrated using serum concentrations and morning void measurements from 39 individuals in North Alabama, all of whom reported the West Morgan-East Lawrence Water Authority as their primary drinking water source in 2010 and 2016. Urine measurements, however, were only available for the 2016 sampling campaign. This dataset was used to refine the model's urine predictions and validate its accuracy against both ATSDR (2016) reference values and Bartell et al. (2010) serum data. Fig. 7 compares the model's predicted serum PFOA concentrations to observed values and evaluates its performance against ATSDR-reported median and mean morning void measurements. The complete parameter dataset, including those used for fitting, has been detailed previously.

3.1.2. Model evaluation

The next step in testing the applicability of the model is to apply it to other studies with "known" intakes and internal concentrations in serum and/or urine. As described in detail earlier, four different studies were used for this purpose. Fig. 8 illustrates the concentrations in the blood of airport workers in Sweden, alongside the corresponding predictions from the PBPK model. The model was run twice, one for one year of exposure and the second for five years of exposure, also capturing the median exposure time that was 2 years. The model performs particularly well in the study of Xu et al. (2020). In the urine, the numerical mean values of the concentrations predicted by the model for the months are presented as blue dots for each month of the sampling period. Although the number of participants providing month-to-month biomonitoring data decreases as urinary PFOA concentrations increase in December and in January, the model demonstrates a very good fit for the urine data as well.

Perez et al. (2012) quantified PFAS (including PFOA) in urine and hair samples from Barcelona residents, although their study did not report participants' consumption patterns or physiological characteristics. To address these limitations, we supplemented their data with: (1) dietary intake estimates from contemporary Spanish population studies (Pérez et al. 2014): 60 ng/kg/day for males, 49 ng/kg/day for females), and (2) Barcelona drinking water PFOA concentrations from the same exposure period (Ericson et al. 2009): 0.85–57.43 ng/L). Our simulations incorporated a mean urinary void frequency of $7 (\pm 2 \text{ SD})$ voids per day, representative of typical human physiology. As shown in Fig. 9, while the model predictions showed some divergence from Perez et al. (2012) measurements - attributable primarily to differences in exposure scenarios and cohort age distributions - the simulated median urine concentrations demonstrated good agreement with the observational data.

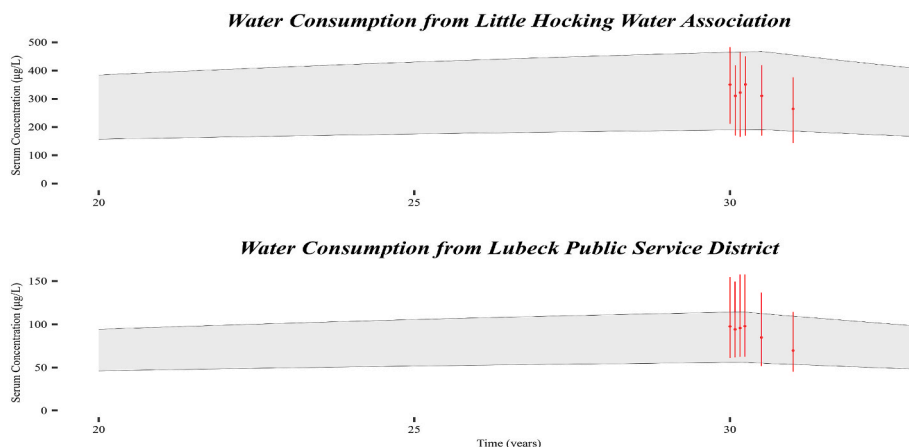


Fig. 6. Comparison of PBPK Model Predictions to Observed Serum PFOA Levels in the Mid-Ohio Valley. The PBPK model-predicted serum PFOA concentrations (black line) are shown alongside measurements from [Bartell et al. \(2010\)](#) (red markers and bounds). Red dots denote mean measured values, with red lines indicating the upper and lower bounds of the observed data. The shaded grey region between the model's upper and lower predictions represents the potential range of individual exposure based on minimum and maximum reported drinking water PFOA concentrations in the study area. This analysis focuses on communities affected by PFOA contamination. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

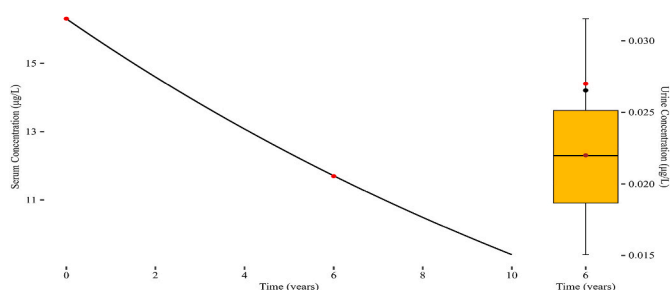


Fig. 7. Comparison of PBPK Model Predictions to Empirical PFOA Data in North Alabama. (Left) Serum PFOA concentrations predicted by the PBPK model (black line) compared to measurements from North Alabama (red dots). The timeline spans 2010 (Year 0) to 2016 (Year 6), with intermediate years proportionally scaled. A 30-year exposure window was assumed before the installation of granular activated carbon filtration system. (Right) Model predictions versus morning void measurements: the brown dot denotes the median value from the [ATSDR \(2016\)](#) study, the red dot represents the measured mean, and the black dot indicates the model-derived mean. All data originates from a PFOA-contaminated region in North Alabama. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In the next study utilized for evaluation purposes, intake data was derived from the work of [Fábrega et al. \(2014\)](#), while [Pérez et al. \(2013\)](#) provided autopsy data for various tissues. It is noteworthy that PFOA concentrations in the brain were below the detection limit and were therefore excluded from further analysis. Kidney measurements were all below the detection limit apart from one; however, the model overpredicts them. This might result from the unknown exposure patterns that each participant had, physiological characteristics as well as exposure characteristics. It is important to highlight that the model reliably predicted concentrations in liver. However, it underperformed in bones compared to observed values, emphasizing the need for further investigation into the biological mechanisms influencing bioaccumulation, not only in bones but across all organs. Comparisons between the model and the autopsy tissue data are shown in [Fig. 10](#).

Finally, the model was evaluated using data from the study by [Lahne et al. \(2024\)](#), where water was also assumed to be the sole source of exposure. Both the lowest and highest concentrations of PFOA in water were considered for the analysis. [Fig. 11](#) demonstrates good agreement between model predictions and biomonitoring data from this study. The

observed underprediction likely reflects our conservative assumption of no additional exposure sources beyond drinking water.

3.1.3. Sensitivity analysis

A global sensitivity analysis was conducted to assess the influence of all parameters on the output variability of the PBPK model. Using the pksensi package, total-order Sobol indices were computed to quantify the contribution of each parameter, accounting for both individual effects and interactions with other variables. The outputs of the analysis are illustrated in [Figs. 12 and 13](#) for the studies of [Bartell et al. \(2010\)](#) and [ATSDR \(2016\)](#). It also includes the fitting data for a better understanding of the impact of the PBPK parameters to the model output. The solid black line represents the median serum concentration trajectory, while the shaded regions depict the percentiles of uncertainty. Specifically, the darkest shaded area corresponds to the interquartile range (25th to 75th percentiles), the medium-shaded area represents the 10th to 90th percentiles, and the lightest shaded region encompasses the full range of predicted values (minimum to maximum). The number of parameters tested in the sensitivity analysis was 149 with the uncertainty indices be presented in the supplementary material, [Table S9](#).

The analysis revealed that blood flow rates were among the most critical determinants of model behaviour. Specifically, gastrointestinal blood flow exhibited the highest sensitivity index, underscoring its significant role in governing systemic and hepatic exposure. Permeability estimates, particularly blood: plasma, GI tract and kidney coefficients, were identified as influential, reflecting their importance in tissue distribution. Liver-specific parameters emerged as another major driver of variability. The protein content in hepatocytes had the highest sensitivity among liver-related parameters, followed by liver tissue density and the liver in-vitro/in-vivo adjustment factor. Transporter kinetics, particularly the affinity of OATP2B1, further modulated hepatic disposition, highlighting the interplay between physiological and biochemical factors in liver clearance. Volume-related parameters, especially those associated with the kidney and pancreas, were also influential, suggesting that tissue composition significantly affects distribution kinetics. The arterial red blood cell volume further contributed to variability, likely due to its role in systemic circulation. The adjustment factor used for kidneys found also to influence significantly the model outputs while the constant of proportionality (used to estimate capillary surface areas) also exhibited measurable influence. Detailed results are presented in the supplementary material ([Table S9](#)).

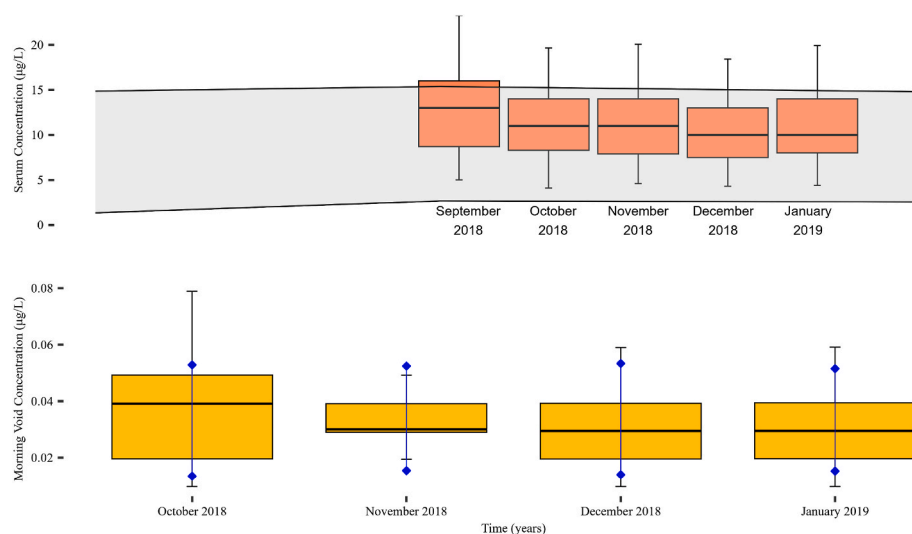


Fig. 8. PBPK Model Performance vs. Human Biomonitoring (HBM) Data in Swedish Airport Workers. (Serum Analysis) Model-predicted serum PFOA concentrations (black lines) compared to HBM data. The lower black line represents a one-year exposure duration, while the upper line corresponds to five years; the shaded region between them reflects intermediate durations. (Urine Analysis) Model-predicted mean urinary PFOA concentrations (blue dots) aligned with boxplots of empirical data. Dots denote mean values from two simulations for corresponding sampling periods. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

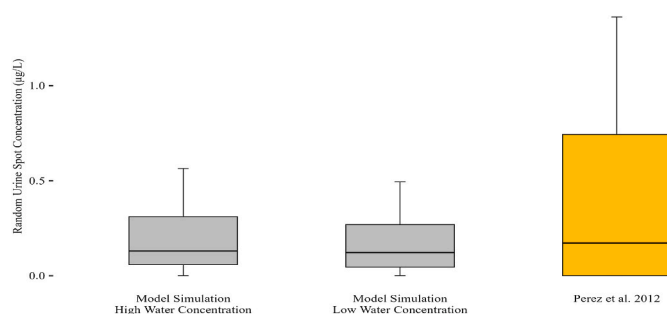


Fig. 9. Random urine spot data against model predictions. This data is unknown if it originated from PFOA contaminated areas.

3.2. External exposure estimates

The development of the model enabled the possibility of running it in reverse. Considering the potential heterogeneity within a population, daily intakes were estimated using HBM data collected from multiple sources (HBM4EU dashboard, NHANES, and literature). For further clarity, the supplementary material (Figs. S1, S2, S3) presents a comparison between the measurements derived from the HBM data and the bioequivalent values calculated earlier. The intakes were normalized per kilogram of body weight to ensure consistency. To facilitate the most accurate understanding of exposure, the safety limits proposed by international organizations were also provided in the following figures while a picture of not log10-transformed intake estimates (Figs. S4, S5, S6) is provided in the supplementary material for robust understandings. Among these, the comparison with the EFSA TWI limit is particularly concerning, as in almost all cases. This observation underscores the need for further extensive research into identifying

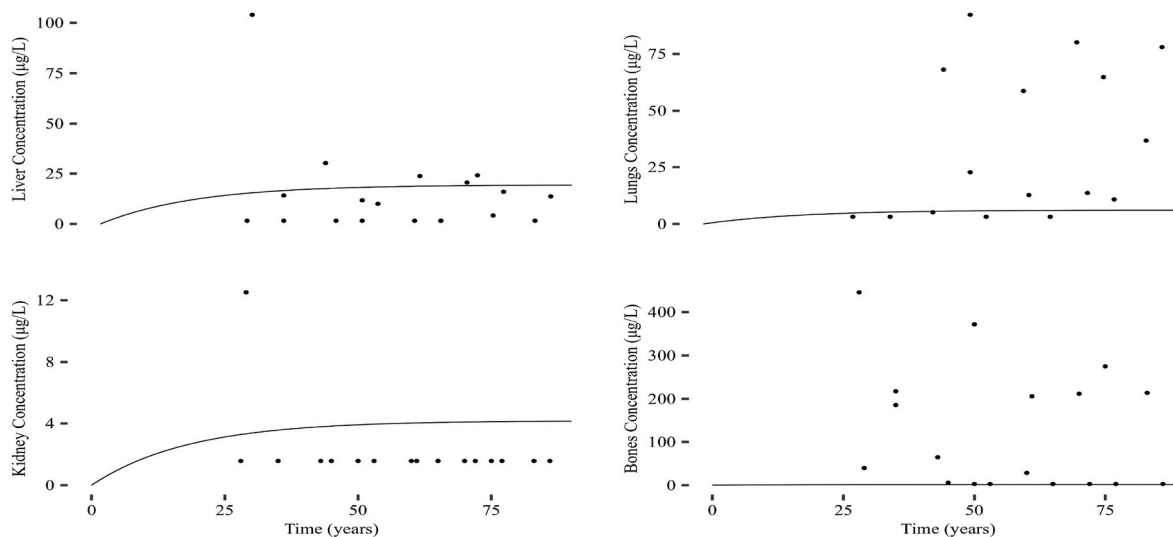


Fig. 10. Performance of the PBPK model (black line) against the autopsy tissue data for individuals from Tarragona. It is not known if any of the individuals sampled were from areas highly contaminated with PFOA.

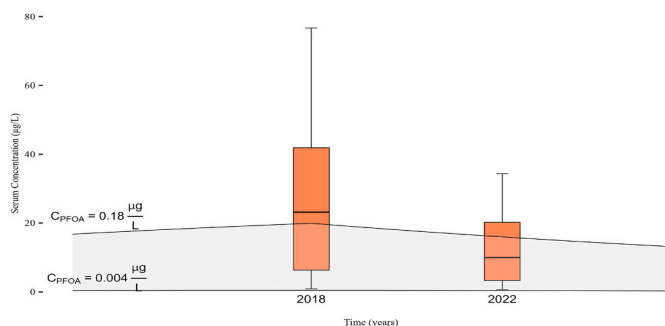


Fig. 11. Performance of the PBPK model (for both exposure concentrations) is compared against the HBM data for the German population exposed to contaminated water. The grey area represents the range of exposure intakes between the two concentrations assessed and other potential contaminated water concentrations.

exposure sources (through targeted exposure modeling and HBM studies) and developing strategies to mitigate them. It is important to consider that a substantial risk to human health is strongly associated with PFAS, including PFOA, even at very low concentrations. Significantly, while the exposure levels remain relatively low compared to the other three reference values, the potential health implications cannot be overlooked.

More specifically, Fig. 14 presents reconstructed intakes originated from HBM4EU dashboard. Concerning the EFSA TWI limits, the calculated intakes appear to be significantly high, highlighting the need to identify and minimize sources of exposure, as previously mentioned. In comparison, the remaining intake levels can be considered relatively safe. These findings indicate a historical decline in PFOA exposure, reflecting the success of minimization policies implemented over the past two decades.

Throughout the timeline, reconstructed intake levels from NHANES studies (Fig. 15) show a general downward trend (as previously), indicating a decline in PFOA exposure in the U.S. population as well. However, the intake estimates consistently exceed the stringent EFSA TWI.

Concerning the literature data as presented in Fig. 16, most reconstructed intake estimates exceed the conservative EFSA TWI and frequently surpass the US EPA TDI, indicating potential health concerns. The study of Hölzer et al. (2008) relies on serum measurements from a contaminated area and as anticipated the intakes are extremely higher compared to the rest. It is important to highlight that temporal trend holds the effectiveness of PFOA restriction policies worldwide.

Finally, based on the reverse dosimetry outputs, conversion factors were estimated for all HBM matrices considered in this study, namely serum, whole blood, and morning voids. For serum, the conversion factor was 0.0014, with a high coefficient of determination ($R^2 = 0.901$), indicating a strong linear relationship between serum concentrations ($\mu\text{g/L}$) and intake ($\mu\text{g/kg/day}$). Serum is a reliable biomarker of exposure regarding PFOA. For whole blood, the conversion factor was 0.0023 with an R^2 of 0.932. Morning voids exhibited a conversion factor of 1.156; however, despite a high coefficient of determination ($R^2 = 0.979$), the limited number of data points ($n = 11$) and the long half-life of PFOA, combined with its low excreted concentrations and other physiological information as previously mentioned, render morning voids an unreliable metric for assessing PFOA exposure. The complete set of conversion equations, along with their corresponding 95 % confidence intervals, are presented below in Equations (8)–(10).

$$\text{Intake} \left(\frac{\mu\text{g}}{\text{kg}_{\text{BW}} \cdot \text{day}} \right) = 0.0014 * C_{\text{Serum}} \left(\frac{\mu\text{g}}{\text{L}} \right), 95\% \text{ CI} : [0.0013, 0.0015], \quad (8)$$

$$\text{Intake} \left(\frac{\mu\text{g}}{\text{kg}_{\text{BW}} \cdot \text{day}} \right) = 0.0023 * C_{\text{Whole Blood}} \left(\frac{\mu\text{g}}{\text{L}} \right), 95\% \text{ CI} : [0.002, 0.003], \quad (9)$$

$$\text{Intake} \left(\frac{\mu\text{g}}{\text{kg}_{\text{BW}} \cdot \text{day}} \right) = 1.156 * C_{\text{Urine}} \left(\frac{\mu\text{g}}{\text{L}} \right), 95\% \text{ CI} : [1.037, 1.276], \quad (10)$$

3.3. Risk characterization

The development of a reliable pharmacokinetic model can also be used to support risk characterization from exposure. The utilization of intake estimates from the previous step (Section 3.2; exposure reconstruction) alongside with human biomonitoring measurements the estimation of risk characterization ratios based on the safety limits established by numerous food safety organizations worldwide (EFSA, 2008a, 2020; FSANZ, 2021; US EPA, 2017). For each data categorization (HBM4EU, NHANES, and literature), two risk ratios are presented: one proportional to the intake and another based on the bioequivalent value calculated earlier (Section 2.9) by applying forward dosimetry. These two values are distinguished by different background colors. Overall, as has already been observed, the risk characterization based on the EFSA TWI remains significantly high.

Fig. 17A presents the results derived from the HBM4EU measurements, showing a clear time trend of risk reduction, proportional to how recent the study is. This trend is also evident in the data originating from the NHANES repository (Fig. 17B) as well as in the literature-derived data (Fig. 17C). Importantly, the study of Hölzer et al. (2008) performed measurements in a highly PFAS-contaminated area in Germany and shows exceptionally high-risk values, as expected from the previous sections. It is also important to highlight that risk estimates are based on concentrations derived from either urine or whole blood samples did not exceed safety limits in all scenarios tested.

3.4. Study limitations

This study also acknowledges numerous limitations. First, the analysis does not utilize real human HBM data distributions, as the distributions from HBM4EU and literature are reconstructed based on aggregated data rather than raw datasets. However, given the aggregated data, the overall picture of reverse dosimetry is a concrete paradigm of aggregated PFOA exposure estimates. Importantly, the estimation of the skewness of the multiple distributions reconstructed for the HBM data might also affect the real picture of the model output. Only the dataset derived from NHANES provided raw biomonitoring data of US citizens to PFOA. However, population overrepresentation issues were not addressed in the framework of this study. Only age and serum concentration of PFOA was considered for reverse dosimetry purposes while the intakes were normalized with the value of 70 kg.

Second, not all transporter kinetics have been fully identified or incorporated into the model, introducing potential gaps in representing the full complexity of PFOA transport mechanisms. For instance, while interactions between PFOA and bone cells are documented in the literature (Koskela et al., 2017), the precise kinetic mechanisms are not well-characterized and thus could not be included in the model. To this end, as revealed by the extensive literature research conducted there are still more toxicity endpoints to be identified regarding the mode of actions of PFOA in all human tissues (Table 1). Moreover, even if they are not identified as the most sensitive parameters for the model, it is important to note that the permeability coefficients employed have not been experimentally validated, potentially affecting the accuracy of predictions regarding substance movement across membranes. Finally, the adjustment factors for transporter kinetics in kidneys and liver are not derived from real human data, introducing a degree of uncertainty in extrapolating results to human physiology.

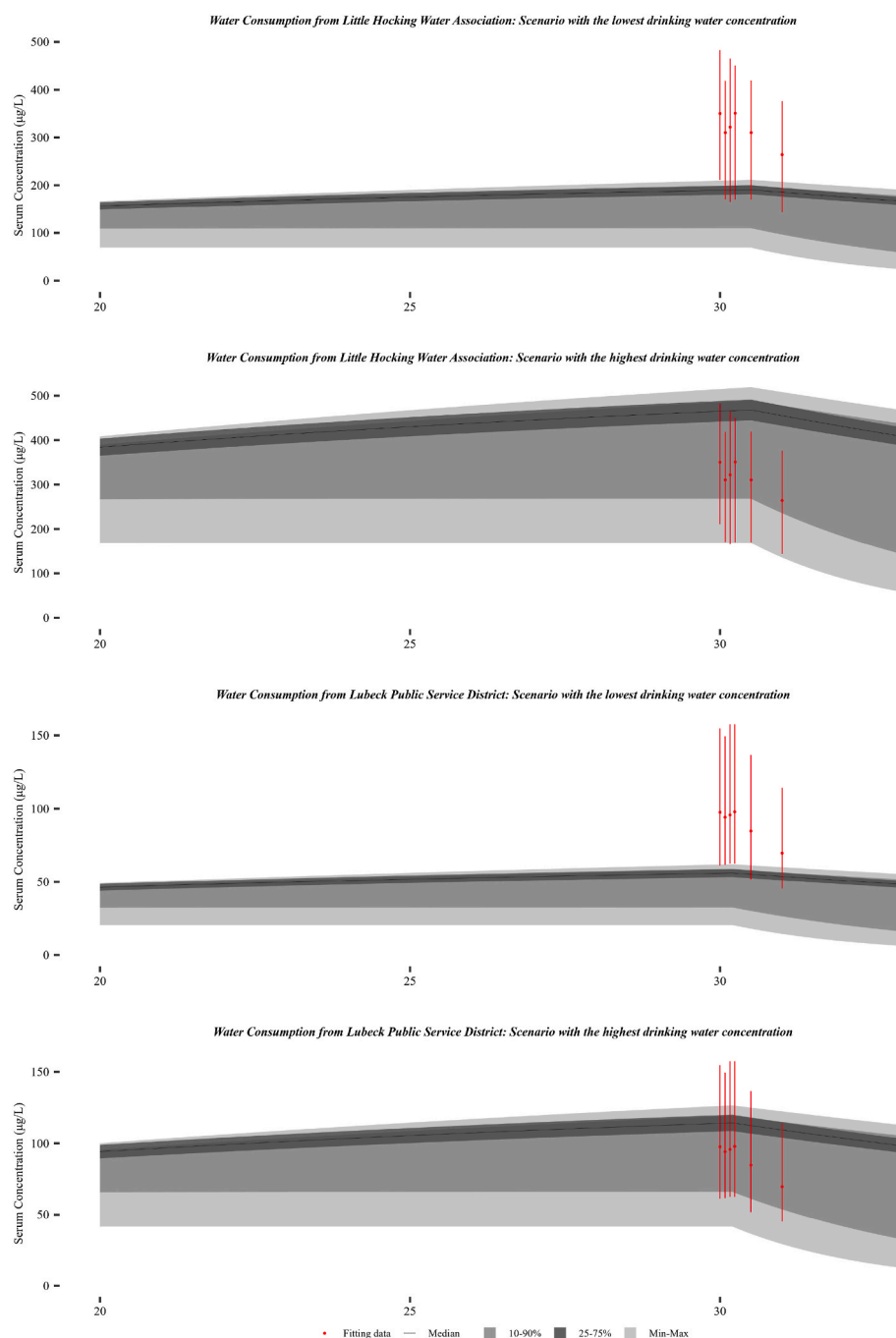


Fig. 12. Sensitivity analysis applied to fitted parameters for Bartell et al. (2010) study.

4. Conclusions

Although the use of PFASs has been more widely banned, they continue to be detected in human biomonitoring (HBM) data and have become firmly established within the food chain. As a result, they are considered a family of chemicals posing very high health risks especially due to their bioaccumulative nature. As outlined previously, PFASs have been linked to numerous diseases in humans, even at low-dose exposures. Consequently, they are considered highly hazardous to human health and are subject to stringent control limits, not only within the EU but also in the US, Australia, and New Zealand.

In this study, a deterministic pharmacokinetic model for PFOA was developed. This model enables the estimation of PFAS concentrations in all major human organs, even at low doses. To further enhance and

evaluate the predictive performance of the current PBPK model, future work should include validation against in vivo time-course pharmacokinetic data. The inverse simulation of the current model allowed for the reconstruction of exposures (bodyweight-normalized intakes) from freely accessible HBM data. Data sources were from HBM4EU dashboard, NHANES, and published literature. It is important to note that, due to the extremely low excretion concentrations of PFOA in urine, to the best of our knowledge, only one study found providing data for this matrix. All other data used was derived from blood-based measurements, especially serum. Additionally, in the present analysis is highlighted the effectiveness of regulatory efforts to reduce PFOA exposures. This is evident in the trend observed, as more recent studies consistently show lower values in the (estimated) exposure levels. This can be drawn from Figs. 3–5 concerning the reduction in PFOA exposure among

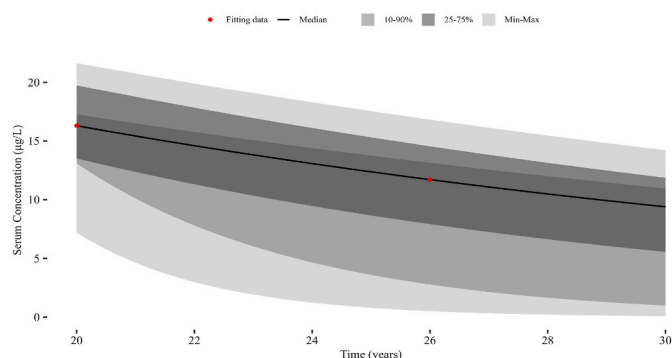


Fig. 13. Sensitivity analysis of PFOA exposure in ATSDR study. The sensitivity analysis output depicts key temporal milestones: Year 20 (2010) marks the installation of water filtration systems, while Year 6 (2016) corresponds to the second biomonitoring measurement in this study. The simulation time before installation was 30 years.

individuals, which positively correlates with the recency of the studies. This indicates that both EU and US regulations and the phasing out of PFOA have been effective. If this trend continues, it is likely that future large cohorts conducted in e.g., Europe, such as those planned under the PARC project (Marx-Stoelting et al., 2023), will reveal even lower exposure values for PFOA and potentially other PFAS.

Using the reconstructed exposure estimates, the risk characterization

ratios based on the recommended limits set by three major organizations across three continents (Europe, the United States, and Australia) was calculated. Unfortunately, according to EFSA's TWI for PFOA, the risk was particularly high, likely due to its more stringent threshold compared to other guidelines. As a result, exposure should be continuously re-evaluated, and targeted studies are needed to identify the primary determinants of PFOA exposure and to inform regulatory measures. In contrast, according to the remaining three limits, participants exhibited lower exposure values, suggesting a comparatively lower risk.

Finally, it is significantly important to highlight that the Chemical Strategy for Sustainability (CSS) relies heavily on PBPK modelling, HBM data, and exposure reconstruction as central components to inform decision-making and their integrative role in advancing chemical risk assessment and ensuring human and environmental safety. PBPK modelling allows for a more accurate and robust prediction of internal exposure levels. HBM data complements this by offering empirical evidence of human exposure to chemicals, serving as a fundamental tool for validating models and identifying exposure trends. Exposure reconstruction further bridges the gap between external exposures and internal dosimetry, allowing for well-established assessments of real-world exposure scenarios and concrete health impact assessments. Together, these tools form a science-based framework that supports the CSS goals of promoting innovation while minimizing chemical risks, eventually contributing to a sustainable and protective regulatory landscape for the human health and the environment.

PFOA Exposure Reconstruction: HBM4EU Dashboard

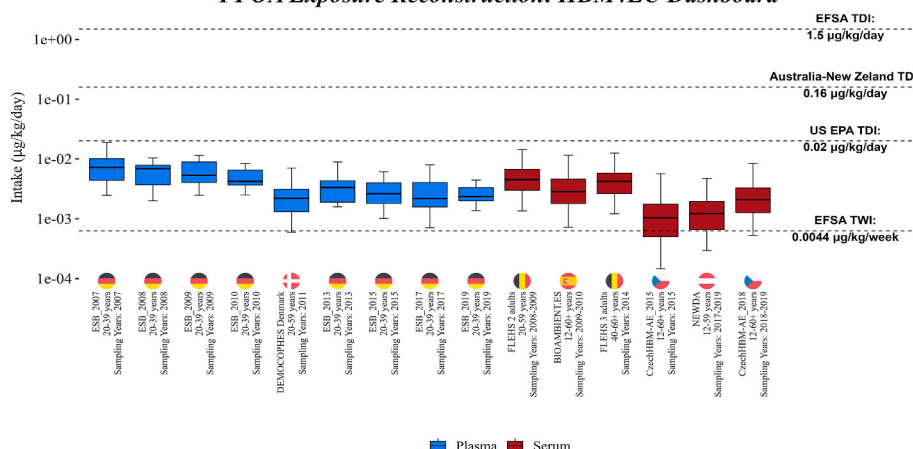


Fig. 14. Presentation of exposure reconstruction output derived from European HBM Dashboard.

PFOA Exposure Reconstruction: NHANES data

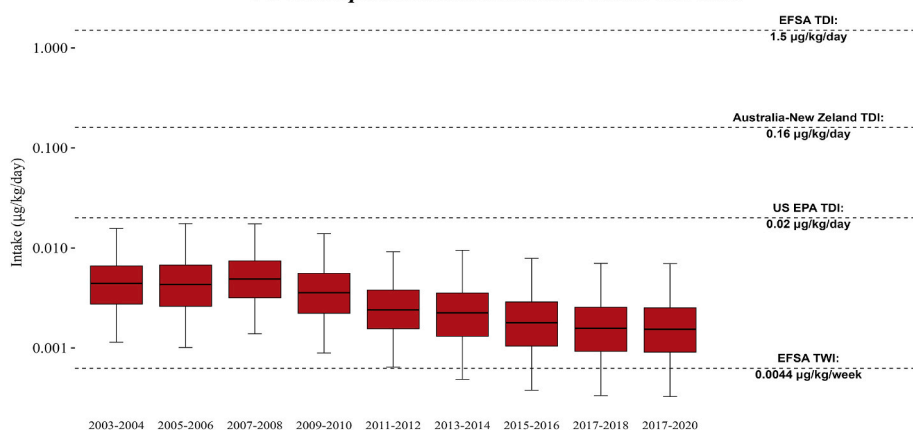


Fig. 15. Presentation of exposure reconstruction output derived from NHANES repository.

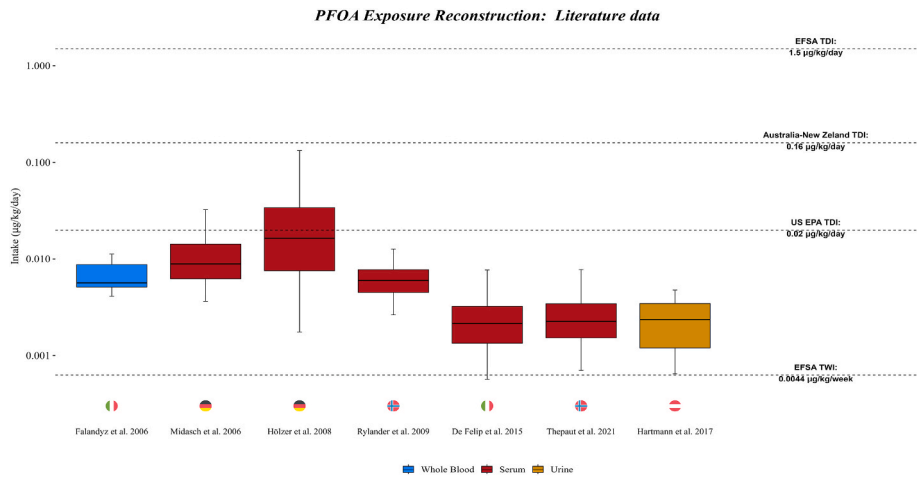


Fig. 16. Presentation of exposure reconstruction output derived from literature search (European data). Measurements in PFOA contaminated reported areas include the studies of [Hölzer et al. \(2008\)](#); while the rest concern measurements on the general population.

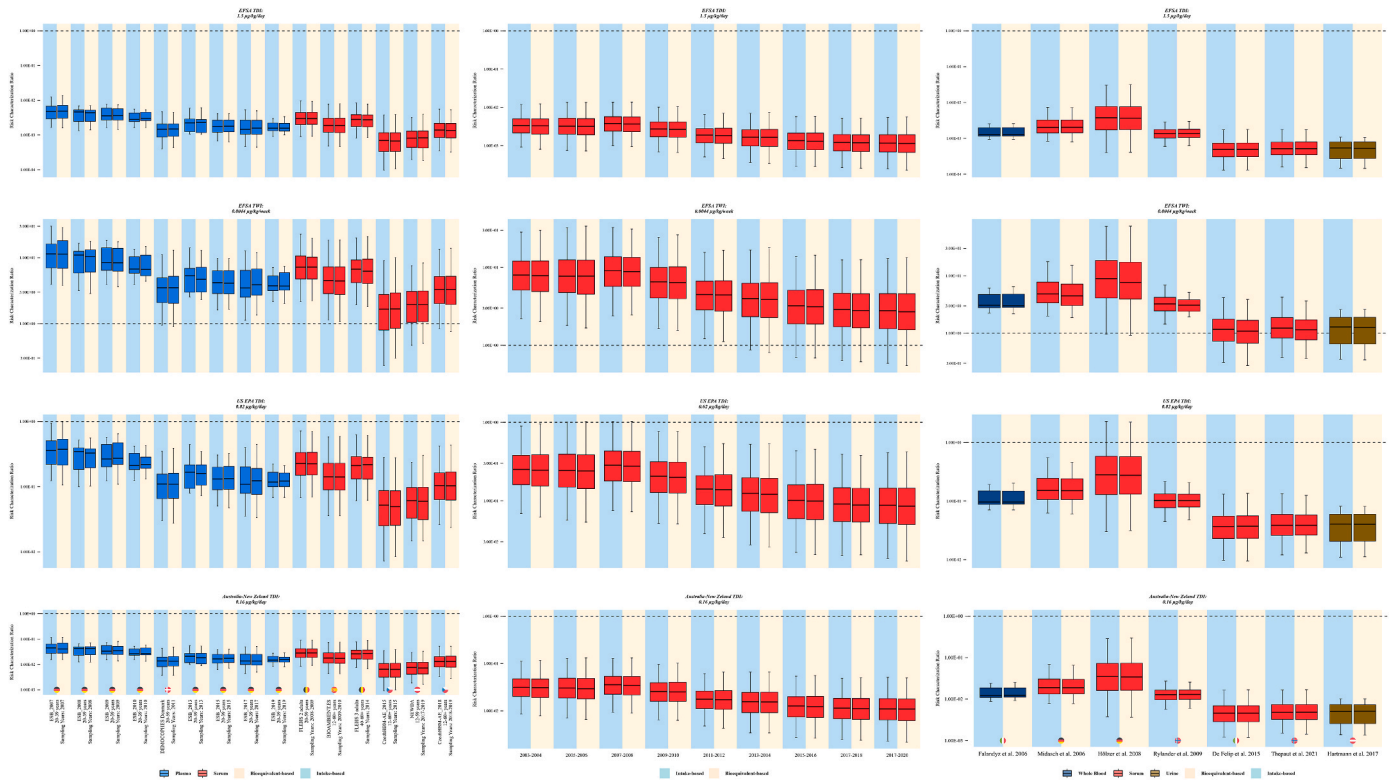


Fig. 17. (A): Risk Characterization Ratios (RCRs) based on reverse and forward dosimetry approaches. RCRs were derived from reverse dosimetry outputs, calculated using an exposure reconstruction methodology as previously described (Sections: 2.7; 3.2). Bioequivalent values were obtained from the HBM4EU dashboard, generated through forward dosimetry. Estimation of both values relied on the sourced from the HBM4EU dashboard. (B): Risk characterization ratios (RCRs) derived from reverse and forward dosimetry outputs with data from the NHANES repository. RCRs were derived from reverse dosimetry outputs, calculated using an exposure reconstruction methodology as previously described (Sections: 2.7; 3.2). Bioequivalent values were generated through forward dosimetry. Estimation of both values relied on the information sourced from the NHANES repository. (C) Risk characterization ratios (RCRs) derived from reverse and forward dosimetry outputs relying on the data collected from the literature. RCRs were derived from reverse dosimetry outputs, calculated using an exposure reconstruction methodology as previously described (Sections: 2.7; 3.2). Bioequivalent values were generated through forward dosimetry. Estimation of both values relied on the information sourced from the NHANES repository.

CRedit authorship contribution statement

Achilleas Karakoltzidis: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Spyros P. Karakitsios:** Writing – review & editing, Writing – original

draft, Supervision, Methodology, Investigation, Conceptualization. **Catherine Gabriel:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Dimosthenis A. Sarigiannis:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2025.121947>.

Data availability

Data obtained from publicly accessible and openly available sources.

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